

Uniform Pluricanonical Iitaka Fibrations

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Abstract

We prove the effective Iitaka fibration conjecture in characteristic zero. For each dimension, one pluricanonical degree defines the Iitaka fibration of every smooth integral projective variety of that dimension and nonnegative Kodaira dimension over an algebraically closed field. The associated sections generate the full Iitaka function field.

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1 Introduction

For a smooth integral projective variety X , the pluricanonical systems organize the birational information carried by its canonical divisor. If some $H^0(X, \mathcal{O}_X(mK_X))$ is nonzero, the maximum of the dimensions of the resulting rational images is the Kodaira dimension $\kappa(X)$. In sufficiently large divisible degrees these maps determine the Iitaka fibration [22]. The question here is whether divisibility and largeness can be prescribed using only $\dim X$.

To specify the required conclusion, let $K(K_X) \subset k(X)$ be the field generated by ratios of nonzero sections in the same positive pluricanonical degree, over all such degrees. This is the function field of the Iitaka base. A complete system *defines the Iitaka fibration* if it is nonempty and its section ratios generate $K(K_X)$. Requiring only an image of dimension $\kappa(X)$ would be weaker: it would allow a proper finite subfield of the Iitaka field.

Theorem 1.1 (Uniform pluricanonical degree). *For each integer $d \geq 1$ there is an integer $m(d) > 0$ such that, over every algebraically closed field of characteristic zero, the complete system $|m(d)K_X|$ defines the Iitaka fibration of every smooth integral projective d -fold X with $\kappa(X) \geq 0$.*

Using the log-abundance input from [LA], this gives an affirmative resolution of the effective Iitaka fibration conjecture [5, Conjecture 1.1]. Every positive multiple of $m(d)$ works as well, by the section-ratio argument in Section 9. Thus the same degree works for all nonnegative Kodaira dimensions. In the general-type case its map is birational. In Kodaira dimension zero the conclusion is nonemptiness in that degree, with image a point. The theorem is existential; no small or practical numerical value of $m(d)$ is asserted.

The proof establishes a second uniform statement at the same time. A pair in the next theorem is normal and integral; its boundary is effective and its log canonical divisor is rational Cartier.

Theorem 1.2 (Uniform log canonical index). *Let $d \geq 0$, and let $\Phi \subset [0, 1] \cap \mathbb{Q}$ satisfy the descending chain condition. There is an integer $a(d, \Phi) > 0$ such that every projective log canonical pair (X, B) of dimension d , with boundary coefficients in Φ and $K_X + B \sim_{\mathbb{Q}} 0$, satisfies*

$$a(d, \Phi)(K_X + B) \sim 0.$$

Here the multiple is an integral principal divisor, not merely a numerically trivial or rationally linearly trivial divisor.

1.1 Context and preceding work

Effective pluricanonical maps are a quantitative part of birational classification. The general-type case asks for a uniform birational degree and was established by Hacon–McKernan, Takayama, and Tsuji [19, 36, 37]. Hacon–McKernan–Xu’s theorem for big log canonical adjoints permits boundary coefficients in a fixed DCC set [20, Theorem 1.3]. The lower-Kodaira-dimensional problem has an additional feature: the general fibre contributes torsion and monodromy data which do not appear in the big case.

Fujino–Mori obtained a uniform pluricanonical degree for threefolds of Kodaira dimension one using their canonical bundle formula [14, Corollary 6.2]. Their denominator argument identifies finite characters on an extended Hodge line and bounds their orders using the middle Betti number of a canonical cover [14, Theorem 3.1 and Sections 3.6–3.8]. Viehweg–Zhang obtained effective results for Iitaka bases of dimension two, with constants controlled by invariants of the general fibre [38, Theorem 0.2]. Birkar–Zhang proved a general effective Iitaka theorem with dependence on the dimension, the least nonvanishing canonical degree of the general fibre, and a middle Betti number of a smooth model of its canonical cover [5, Theorem 1.2]. Their effective birationality theorem for polarized adjoints is the final effective input used here [5, Theorem 1.3]. The present argument supplies the required denominator control without assuming a uniform bound for those middle Betti numbers.

The index problem has its own inductive structure. Xu relates zero-boundary klt index bounds to bounds with boundary [39, Theorem 1.8]. Birkar proves the index theorem for rationally connected klt Calabi–Yau pairs in every fixed dimension [4, Corollary 1.8]; his boundedness theorem for generalized Calabi–Yau pairs also controls torsion on rationally connected bases [4, Theorem 1.7]. We combine these results with an arithmetic Stein-degree theorem [SD, Main Theorem] to pass from klt pairs to normal lc pairs without an index theorem for nonnormal pairs.

1.2 Inputs and contributions

One major input must be distinguished from standard minimal-model technology. We use the good-log-minimal-model theorem of the companion *Log abundance in characteristic zero* [LA, Theorem 11.1], stated precisely in Theorem 2.1. It includes nonvanishing and is used in all dimensions. It is not being inferred from finite generation for big adjoints. The other companion inputs are the chain and tracking lemmas in [U4], and the arithmetic Stein-degree bound in [SD]. The arguments of [R4] provide the preceding fourfold treatment of relative denominators and rationally connected torsion.

The proof extends the canonical-index method of [U4] to arbitrary dimension. There are two substantial extensions. First, we prove a relative denominator bound by decomposing the geometric generic fibre, degenerating its individual factors, and controlling their volume-form characters on normal components of a dlt special fibre. This separates the potentially unbounded ramification of semistable reduction from the bounded characters that enter the moduli divisor. The argument uses the absence of horizontal boundary and only lower-dimensional normal lc index bounds. It is therefore a reusable relative statement, not merely the final step of the Iitaka theorem.

Second, the small-volume argument in [U4] is extended to Iitaka bases of arbitrary dimension. Weak positivity supplies a fibre-to-total volume comparison, while flattening ensures that codimension-two losses on the base create no divisorial poles on the original smooth model. This gives scalar bounds for section orders and Seshadri constants. We then give the dimension-dependent calculations in the Frobenius diagonal argument and in the compatible-chain argument. Those constructions are adaptations of [U4]; the new claim is their use in the all-dimensional induction, not a new origin for the constructions themselves.

1.3 The induction

We prove Theorems 1.1 and 1.2 simultaneously. Fix n and suppose both are known in smaller dimensions. The relative step starts with a contraction $f : X \rightarrow Z$ from a klt n -fold, with $\dim Z > 0$ and K_X rationally linearly pulled back from Z . The canonical bundle formula writes

$$K_X \sim_{\mathbb{Q}} f^*(K_Z + B_Z + M_Z),$$

where B_Z is the discriminant divisor and M_Z is the moduli divisor. Section 4 bounds a Cartier multiple of the moduli divisor on a birational base model using only lower-dimensional normal lc indices. When $K_X \sim_{\mathbb{Q}} 0$ and the base has a rationally connected smooth resolution, this denominator bound combines with torsion control on the base to bound the canonical index of X .

Section 5 then reduces failure of the zero-boundary klt index bound to terminal varieties V with canonical indices $r \rightarrow \infty$ and countable birational groups. Their canonical cyclic covers $\pi : Y \rightarrow V$ have degree r and deck group G . Every G -equivariant rational fibration of Y with positive-dimensional base and fibre has general-type smooth geometric generic fibre. We call such fibrations intermediate. The corresponding assertion holds for intermediate rational fibrations of V .

Choose an ample rational Cartier divisor L on V with $1 \leq L^n \leq 2$. For an ample rational Cartier divisor P , write $\varepsilon(P)$ for its very general Seshadri constant: the infimum of $P \cdot C / \text{mult}_x C$ over integral curves through a very general point x . Section 6 uses lower-dimensional effective Iitaka fibrations to bound section orders on V and bound $\varepsilon(L)$ below uniformly. The pullback polarization has $(\pi^*L)^n = rL^n$.

For $t \geq 2$, let $Z_t = Y^t / G_{\text{diag}}$, where G acts by the same deck transformation in every slot. Let $\theta_t : Z_t \rightarrow V^t$ be the induced finite map and set $P_t = \theta_t^*(\sum_{i=1}^t \text{pr}_i^* L)$. Together, the scalar estimates

and the diagonal argument of Section 7 give

$$\varepsilon(\pi^*L) \geq cr^{1/n}, \quad \varepsilon(P_t) \leq Ct^2,$$

with $c, C > 0$ depending only on n . The second bound supplies curves of bounded P_t -degree divided by marked multiplicity.

Section 8 arranges these curves into chains compatible with coordinate projections. The resulting positive-dimensional leaves project generically finitely onto their images in each coordinate, and comparison of their projection degrees gives a degree-one coordinate-forgetting map. Rational recovery of its missing coordinate turns local flows into uncountably many birational self-maps of V , a contradiction. This bounds the zero-boundary klt index. The normal lc reduction in Section 3 then proves Theorem 1.2 in dimension n . Finally, Section 9 reuses the denominator bound with polarized effective birationality to prove Theorem 1.1 in that dimension. Figure 1 records the dependencies after the induction statements have been formally introduced.

2 Conventions and the induction hypotheses

Until the final field-descent argument we work over $\mathbb{C}$. Varieties are integral and projective unless another setting is stated. We use the standard discrepancy conventions for klt, dlt, and lc pairs [25]. All boundaries in the proof are rational. A *rational Cartier divisor* means a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor. Canonical divisors are chosen compatibly when forms are pulled back or residues are taken. On a normal variety, sections of an integral Weil divisor always mean sections of its rank-one reflexive divisorial sheaf.

For rational divisors D, E we write $D \preceq E$ if $E - D$ is rationally linearly equivalent to an effective divisor. An index bound is a *common trivializing multiple*: a single integer kills all the actual divisor classes under consideration and clears their coefficients. A bound on the least possible integers implies a common multiple by taking their least common multiple.

A *contraction* is a projective surjective morphism $f : X \rightarrow Z$ of normal varieties such that $f_*\mathcal{O}_X = \mathcal{O}_Z$. Its generic fibre is geometrically integral in characteristic zero. A rational fibration is understood through a relatively algebraically closed subfield of the total function field; after resolution and Stein factorization it has connected fibres. A smooth geometric generic model is a smooth projective model of the geometric generic fibre.

For generalized pairs, the nef data consist of a nef rational Cartier divisor on a higher birational model, together with its compatible pullbacks. A moduli b-divisor is *b-nef* and *b-Cartier* if it is the Cartier closure of a nef divisor on one such model. The generalized discrepancies are computed by pulling back the full adjoint, including these nef data. We use the conventions of [5].

Theorem 2.1 (Good minimal models; [LA]). *Let (X, B) be a projective lc pair over $\mathbb{C}$ with effective rational boundary and pseudo-effective $K_X + B$. Then (X, B) has a good log minimal model. In particular, its adjoint is nonvanishing, and on the good model the adjoint is semiample.*

This is the rational-boundary case of [LA, Theorem 11.1]. We also use the standard klt MMP and its discrepancy comparisons [2]. For klt pairs with a good minimal model, an MMP with scaling can be chosen terminating at such a model. We will specify relative or equivariant uses when they arise; they do not require a separate unmentioned abundance assumption.

Notation 2.2 (Induction statements). Let I_n denote Theorem 1.1 in dimension n , and let L_n denote Theorem 1.2 in dimension n , for every rational DCC coefficient set. Let K_n denote its zero-boundary klt case: there is a common integer $a_n > 0$ with $a_n K_X \sim 0$ for all projective klt n -folds satisfying $K_X \sim_{\mathbb{Q}} 0$.

The dimension-zero statements hold with degree one. In a fixed induction step we assume I_j, L_j for all $j < n$. By global ACC [20, Theorem 1.5], the coefficients of numerically trivial lc pairs of fixed dimension with coefficients in a fixed DCC set belong to a finite subset. Thus proving L_n for every finite rational coefficient set proves its stated DCC form. This observation also applies to the different coefficients that occur in adjunction.

The dependency diagram below distinguishes K_n from L_n . In particular, the proof of the moduli denominator uses only L_j with $j < n$, not the dimension- n index statement being established.

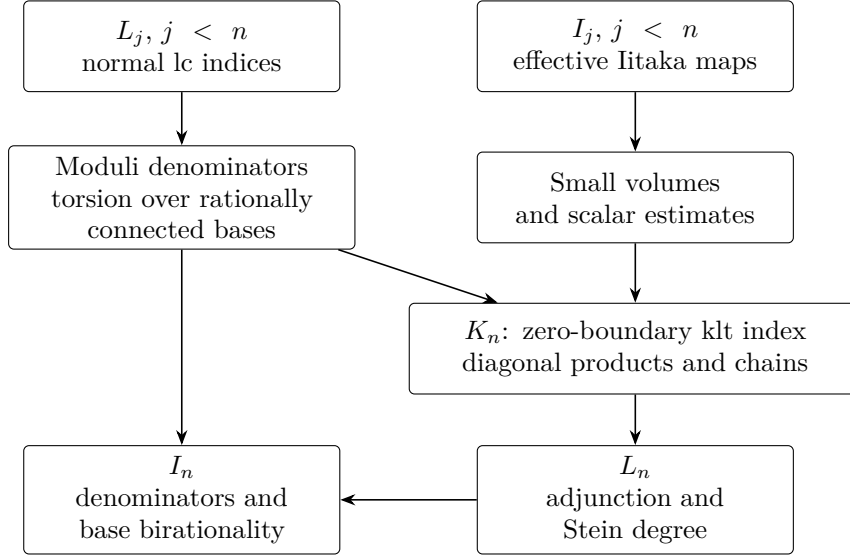


Figure 1: One induction step. The left-hand route supplies the relative index control used in reducing K_n ; the right-hand route supplies its section-order estimates. After K_n is proved, adjunction and Stein degree give L_n , and the denominator bound also feeds the final proof of I_n , as shown by the direct left-hand arrow. Arrows record the main dependencies, not morphisms of varieties.

2.1 Geometric generic fibres and fields of definition

All finite collections of varieties, maps, divisors, and rational forms can be defined over a finitely generated field of characteristic zero. Algebraically closed extensions preserve the singularity conditions, dimensions of section spaces, Cartierness, and triviality of the line bundles used here. For singularities one may spread a log resolution and its discrepancy calculation; for sections and trivializations one uses flat base change and faithful flatness. Consequently the induction hypotheses apply to geometric generic fibres, by comparison with an algebraically closed field embedded in $\mathbb{C}$.

We occasionally need very general points to satisfy countably many section-order conditions simultaneously. One may make these choices over $\mathbb{C}$, or use points geometric generic over a countable field of definition for the varieties and all the systems involved. Moving members below are taken at geometric generic parameters. Dominant extensions needed to define models or maps do not specialize a previously chosen generic parameter.

We record one descent observation used repeatedly. If F is a geometrically integral normal projective variety over a field K and a given reflexive pluricanonical sheaf becomes the trivial line bundle over $\bar{K}$, then it is already a line bundle over K and is trivial there. Local freeness descends faithfully flatly. Base change identifies its space of sections with a one-dimensional space after extension, and any nonzero section over K becomes a generator, hence is a generator before

extension as well. The same argument applies to a specified integral log pluricanonical divisor.

3 Normal log canonical indices and finite covers

Two preliminary arguments will keep the induction within normal varieties. The first recovers an lc index from the index of one normalized boundary component and its Stein degree. The second separates finite actions on the factors of a quasi-étale product cover. Throughout this section, the inductive assertions I_j, L_j are available for $j < n$; the assertion K_n is used only in Proposition 3.2.

3.1 Descent from a normalized boundary component

We begin with an elementary descent observation. A rational q -canonical form means a nonzero rational section of the q th tensor power of the canonical line at the generic point. Its divisor is defined on a normal variety by orders at prime divisors.

Lemma 3.1 (Descent of a vertical principal divisor). *Let $f : X \rightarrow Z$ be a contraction of normal projective varieties over a field of characteristic zero. A rational function on X whose divisor has no horizontal component belongs to the subfield $k(Z) \subset k(X)$. Suppose, in addition, that (X, B) is a rational pair with $K_X + B \sim_{\mathbb{Q}} 0$, and that $q(K_{X_\eta} + B_\eta) \sim 0$ on the generic fibre. There are a rational q -canonical form s and a rational Cartier divisor $D \sim_{\mathbb{Q}} 0$ on Z such that*

$$\operatorname{div}(s) + qB = f^*D. \quad (3.1)$$

Proof. The generic fibre X_η is normal and projective, with $H^0(X_\eta, \mathcal{O}_{X_\eta}) = k(Z)$. A rational function with vertical divisor restricts to a function with neither zeros nor poles at any prime divisor of X_η . Normality makes it regular, and hence a member of $k(Z)$.

For the second assertion, extend a trivializing rational form on X_η to a rational q -canonical form s on X , using any rational top form on Z to identify relative and absolute canonical lines. The divisor $H = \operatorname{div}(s) + qB$ is vertical. Choose $a > 0$ and a rational aq -canonical form τ with $\operatorname{div}(\tau) + aqB = 0$. The function $s^{\otimes a}/\tau$ has divisor aH , so the first assertion writes it as f^*h for some $h \in k(Z)^*$. Thus $H = f^*((1/a)\operatorname{div}(h))$, as required. $\square$

Proposition 3.2 (Reduction to the zero-boundary klt index). *Assume L_j for every $j < n$ and K_n . Then L_n holds: for every finite set $\Gamma \subset [0, 1] \cap \mathbb{Q}$ there is an integer $c(n, \Gamma) > 0$ such that*

$$c(n, \Gamma)(K_X + B) \sim 0$$

for every normal projective lc n -fold pair (X, B) with coefficients in Γ and $K_X + B \sim_{\mathbb{Q}} 0$.

Proof. The assumptions give the zero-boundary klt index bounds through dimension n . Xu's klt induction theorem [39, Theorem 1.8] therefore supplies the assertion when (X, B) is klt. We prove the non-klt case without an slc index statement.

A Mori fibre space and a horizontal lc divisor. Take a crepant $\mathbb{Q}$ -factorial dlt modification, and enlarge Γ by 1. Write $T = \lfloor B \rfloor \neq 0$. For sufficiently small rational $\epsilon > 0$, the pair $(X, B - \epsilon T)$ is klt and

$$K_X + B - \epsilon T \sim_{\mathbb{Q}} -\epsilon T$$

is not pseudo-effective: its intersection with the $(n-1)$ st power of an ample divisor is negative. The klt minimal model program [2, Corollary 1.3.3] ends in a Mori fibre space $f : X' \rightarrow Z$.

The full log canonical divisor remains crepant through this program. Indeed, fix compatible canonical divisors and write $r(K_X + B) = \operatorname{div}(h)$. Divisorial contractions push forward this same

identity, and flips preserve it under their codimension-one identification. On a common resolution, both pullbacks equal $r^{-1} \operatorname{div}(h)$. Thus the full pairs on all models are lc, even though they need not be dlt. The exact order of their log canonical divisors is unchanged. We henceforth write (X, B) for the last model.

If $\dim Z = 0$, the klt log Fano pair $(X, B - \epsilon T)$ shows that X is of Fano type. The bounded complement theorem [3, Theorem 1.7], with a fixed hyperstandard set containing Γ , gives a bounded integer c and a complement $B^+ \geq B$ with $c(K_X + B^+) \sim 0$. The effective divisor $B^+ - B$ is numerically trivial, hence zero by intersection with an ample class. This settles that case.

Suppose $\dim Z > 0$. On the Mori fibre space the surviving transform of T is relatively ample, since ϵT is relatively $\mathbb{Q}$ -linearly equivalent to $-(K_X + B - \epsilon T)$. In particular, some prime component S of T dominates Z . Let $\nu : S^\nu \rightarrow S$ be its normalization. Divisorial adjunction gives

$$(K_X + B)|_{S^\nu} = K_{S^\nu} + \Theta. \quad (3.2)$$

Here (S^ν, Θ) is a normal lc pair and Θ is effective; its coefficients belong to the rational DCC set of differents determined by Γ [27]. Its adjoint is $\mathbb{Q}$ -linearly trivial. The lower-dimensional assertion L_{n-1} , together with global ACC as in the conventions, therefore bounds a trivializing multiple of $K_{S^\nu} + \Theta$.

A common form degree. The geometric generic fibre of f is a normal projective lc pair of dimension less than n , with the same coefficient set. The assertions L_j , $j < n$, bound a trivializing multiple there, and linear triviality descends in that same degree to $k(Z)$. To see the last point, the corresponding divisorial sheaf becomes an invertible trivial sheaf after algebraic closure. Invertibility descends faithfully flatly. Flat base change for global sections then gives a one-dimensional space over $k(Z)$, and its evaluation map is an isomorphism because it becomes one after the field extension.

Choose a common integer q , depending only on n and Γ , which clears Γ and trivializes both this fibre adjoint and $K_{S^\nu} + \Theta$. Lemma 3.1 gives (3.1), with $D \sim_{\mathbb{Q}} 0$. At the generic point of S , the form s has precisely its logarithmic pole of order q . Its q -fold residue is a nonzero rational q -canonical form v on S^ν , and adjunction gives the equality of actual divisors

$$\operatorname{div}(v) + q\Theta = (f \circ \nu)^* D. \quad (3.3)$$

For this equality one may first take a sufficiently divisible tensor power and apply the usual pluricanonical adjunction isomorphism; dividing the resulting equality of divisorial orders gives (3.3). No Cartier assumption in degree q along the whole of S is needed.

Stein factorization and norm. Choose a trivializing q -log canonical form v_0 on (S^ν, Θ) . Then $a = v/v_0 \in k(S)^*$ satisfies

$$\operatorname{div}(a) = (f \circ \nu)^* D.$$

Factor the map as $S^\nu \xrightarrow{g} Y \xrightarrow{h} Z$, where g is a contraction and h is finite, with Y normal. The first part of Lemma 3.1 gives $a \in k(Y)^*$. Injectivity of pullback for rational Cartier divisors under the surjective map g yields

$$h^* D = \operatorname{div}_Y(a).$$

One can check this injectivity at a prime divisor of Y , where a nonzero order remains nonzero at a prime divisor above it. The norm identity for the finite morphism h now gives

$$(\deg h) D = \operatorname{div}_Z(N_{k(Y)/k(Z)} a). \quad (3.4)$$

It remains to bound $\deg h$, not the degree of any gluing cover. Apply the arithmetic Stein-degree theorem [SD, Theorem 1.1] to (X_η, B_η) over $k(Z)$, with its prime coefficient-one divisor S_η and

parameter $t = 1$. This fibre is normal, integral and projective, and its field of global functions is $k(Z)$. The relative algebraic closure of $k(Z)$ in $k(S_\eta)$ is exactly $k(Y)$. Hence $\deg h$ is bounded in terms of the fibre dimension, and therefore in terms of n .

Equations (3.1) and (3.4) show that $q(\deg h)(K_X + B)$ is principal. Taking the least common multiple of the finitely many possible degrees, and then a common multiple with the klt and base-point cases, proves the proposition. The earlier crepant comparisons return the same multiple to the original pair. $\square$

The normality restriction in Proposition 3.2 is important. The argument uses a single normalized component and takes a field norm; it does not trivialize the log canonical sheaf on the possibly reducible scheme $[B]$.

The same index hypothesis also controls the action of finite automorphisms on log volume forms. The following consequence will be used on components of degenerating fibres in Section 4.

Lemma 3.3 (Finite-order log volume characters). *Assume L_j in a fixed dimension j . There is an integer $c_j > 0$ with the following property. Let (V, E) be a normal projective lc j -fold pair, with E reduced, and let ω be a rational top form satisfying $\operatorname{div}(\omega) + E = 0$. If γ is a finite-order automorphism of the pair and $\gamma^*\omega = \lambda\omega$, then $\lambda^{c_j} = 1$.*

Proof. Form the quotient $\pi : V \rightarrow W = V/\langle\gamma\rangle$. Define the boundary E_W by the finite adjunction formula

$$K_V + E = \pi^*(K_W + E_W).$$

At a prime divisor with ramification index e , its coefficient is $1 - 1/e$ if the upstairs coefficient is zero, and 1 if the upstairs coefficient is one. Thus the coefficients belong to the fixed rational DCC set $\{1 - 1/e : e \in \mathbb{N}\} \cup \{1\}$. The finite discrepancy formula makes (W, E_W) lc [27, Proposition 5.20]. A power of ω invariant under the finite cyclic group descends to a rational pluricanonical form downstairs; the ramification formula shows that it trivializes a multiple of $K_W + E_W$. This also establishes its $\mathbb{Q}$ -Cartierness and $\mathbb{Q}$ -linear triviality.

The assertion L_j and global ACC supply a uniform c_j for this DCC set such that $c_j(K_W + E_W) \sim 0$. Pull back a trivializing log pluricanonical form. It is an invariant generator upstairs, necessarily a scalar multiple of $\omega^{\otimes c_j}$, since V is projective and integral. Therefore $\lambda^{c_j} = 1$. For $j = 0$ the same conclusion holds with $c_0 = 1$. $\square$

3.2 Quasi-étale covers and factor actions

A finite map of normal varieties is *quasi-étale* if it is étale outside a set of codimension at least two. We write $\Omega_V^{[p]}$ for the reflexive extension of p -forms from the smooth locus. The elementary character criterion below records that a bound on a volume character controls the exact global canonical index, including Cartierness.

Lemma 3.4 (Canonical index and the volume character). *Let $\pi : R \rightarrow X$ be a finite Galois quasi-étale cover of normal projective varieties, with $K_R \sim 0$. Choose a generating reflexive top form ω on R , and write $g^*\omega = \chi(g)\omega$ for the Galois action. Then*

$$mK_X \sim 0 \iff \chi^m = 1.$$

Proof. If $\omega^{\otimes m}$ is invariant, it descends at the function field level to a rational m -canonical form on X . Quasi-étaleness means that its divisor has order zero at every prime divisor of X . It therefore trivializes $\mathcal{O}_X(mK_X)$ as a divisorial sheaf, so mK_X is principal. Conversely, a trivializing form downstairs pulls back to a scalar multiple of $\omega^{\otimes m}$ and is invariant. $\square$

For a projective klt variety X with $K_X \sim_{\mathbb{Q}} 0$, the cyclic cover attached to a primitive trivialization of its canonical multiple is connected and quasi-étale. It has Cartier trivial canonical divisor and is klt by finite adjunction, hence canonical: its discrepancies are integers greater than -1 . The singular Beauville–Bogomolov decomposition theorem then supplies a finite quasi-étale cover

$$P = A \times \prod_{i=1}^r B_i \longrightarrow X. \quad (3.5)$$

Here A is abelian and each B_i is an irreducible Calabi–Yau or irreducible holomorphic symplectic variety in the singular sense [21, Definition 1.4 and Theorem 1.5]. This theorem combines the splitting and holonomy results of Druel and Greb–Guenancia–Kebekus [11, 16].

We use the following part of the definition of these nonabelian blocks. They and all their connected normal finite quasi-étale covers are canonical with trivial canonical divisor. Their reflexive form algebras are generated by a top form in the Calabi–Yau case, or by a generically nondegenerate two-form in the symplectic case. In particular they have no reflexive one-forms and no reflexive forms in degree one below their dimension. Contraction with a volume form also gives the absence of vector fields.

The cover $P \rightarrow X$ need not be Galois, and its Galois closure need not remain a product. The next lemma is the comparison needed to use the factors nonetheless. The argument follows the product direction construction in [U4, Section 4.1]; we include the regularity and semilinear details required here.

Lemma 3.5 (Equivariant comparison with the factors). *Let k be a field of characteristic zero and let $P = \prod_{i=1}^t B_i$ be a product of geometrically integral projective factors whose geometric fibres are positive dimensional blocks as above; include the abelian part, when present, as a single factor. Let $R \rightarrow P$ be a geometrically integral normal finite quasi-étale cover, and put $d = \dim R$. If a finite group G acts on R , then there is a subgroup G_0 of index at most $d!$ and connected finite quasi-étale covers $R_i \rightarrow B_i$ with the following properties:*

1. $R \rightarrow \prod_i R_i$ is finite and quasi-étale;
2. G_0 acts regularly on each R_i , and the projections $R \rightarrow R_i$ are equivariant;
3. a generating volume form on R is proportional to the wedge product of the pulled-back volume generators on the R_i .

The conclusions also hold for finite semilinear actions, interpreting a factor transformation as an isomorphism to the corresponding field-conjugate variety. In particular the $d!$ th power of any finite-order transformation preserves all factor fields.

Proof. Intrinsic preservation of the product directions. Pullback on the common smooth big open, followed by reflexive extension, gives

$$\mathcal{T}_R = \bigoplus_{i=1}^t \mathcal{E}_i. \quad (3.6)$$

There are no global homomorphisms between two distinct summands. This can be checked after extending k to an algebraic closure. At least one of the corresponding factors is nonabelian, say B_i . Fix general smooth points in all complementary factors. The resulting slices of R are normal and finite over B_i : generic flatness and geometric normality in characteristic zero allow this after shrinking the complementary parameter space. The locus omitted from the quasi-étale map has codimension at least two on a general slice. Each connected component of the slice is consequently a connected normal quasi-étale cover of B_i .

On the smooth big open of that component, $\mathcal{E}_i$ is its tangent sheaf, while the complementary summands are trivial. A homomorphism in either direction gives a vector field or a one-form, which extends reflexively and vanishes by the block properties. These slices cover a dense open of R , proving the claimed vanishing.

The block projections in (3.6) are therefore central idempotents of the finite-dimensional algebra $\text{End}_R(\mathcal{T}_R)$. Its primitive central idempotents give nonzero direct summands of positive generic rank, so there are at most d of them. The action of G permutes this finite set. Its kernel G_0 has index at most $d!$ and fixes each block projection, since a central idempotent is a sum of primitive central idempotents. This reasoning is unchanged for semilinear algebra automorphisms. In particular, the idempotent argument is carried out over k itself; it does not require extending the finite group action to an algebraic closure.

Recovering the finite factor covers. Let F_i be the relative algebraic closure of $k(B_i)$ in $k(R)$, and let R_i be the normalization of B_i in F_i . Equivalently, $R \rightarrow R_i \rightarrow B_i$ is the Stein factorization of the projection. The intermediate fields F_i are regular over k , because $k(R)$ is regular over k ; hence the R_i are geometrically integral. Stein factorization commutes here with algebraic closure: the extensions $k(R)/F_i$ are regular in characteristic zero. At the generic point, the complementary directions in (3.6) span $\text{Der}_{k(B_i)} k(R)$. For a finitely generated extension in characteristic zero, the common constants of these derivations are exactly the elements algebraic over $k(B_i)$. Consequently G_0 preserves F_i and induces birational factor transformations.

The extension obtained by adjoining the other product factors to $k(B_i)$ is regular, and hence is linearly disjoint from the finite extension $F_i/k(B_i)$. Thus the function field of the normal product $R_i \times \prod_{j \neq i} B_j$ is intermediate between $k(P)$ and $k(R)$. Normalization gives finite maps

$$R \longrightarrow R_i \times \prod_{j \neq i} B_j \longrightarrow P.$$

Ramification over a divisor of B_i would persist in this product and then at a prolongation to $k(R)$, contradicting quasi-étaleness. Thus $R_i \rightarrow B_i$ is quasi-étale. The product map $R \rightarrow \prod_i R_i$ is proper with finite fibres, because its composite to P is finite. Its image has dimension d , equal to the dimension of the integral target product, so it is surjective and finite. The same ramification argument shows it is quasi-étale. In particular every fibre of $R \rightarrow R_i$ has dimension $d - \dim R_i$.

Why the factor transformations are regular. Fix $g \in G_0$, and let Γ_i be the closed graph of its induced birational transformation on R_i . The two morphisms from R to R_i give a surjection $R \rightarrow \Gamma_i$. All its fibres have dimension at least $d - \dim R_i$. If the first projection $\Gamma_i \rightarrow R_i$ had a positive-dimensional fibre, its preimage in R would have dimension strictly larger than $d - \dim R_i$, contradicting the preceding equidimensionality. Thus this projection is finite and birational. Normality of R_i makes it an isomorphism. Applying the same argument to g^{-1} proves that the induced transformation is an automorphism. For a semilinear transformation, replace the target by its field conjugate; the argument is identical.

Finally, each geometric R_i is a quasi-étale cover of its geometric block. It is canonical and has trivial canonical divisor. These properties descend to k ; linear triviality descends by the same one-dimensional-section argument used in Proposition 3.2. The product of its volume generators pulls back under the finite quasi-étale product map to a generator of K_R . Any two generators differ by a nonzero constant. This proves all three assertions. $\square$

The semilinear version applies to families over a curve as follows. After a finite extension of the curve function field, define the geometric product cover and its blocks. Take the normal Galois closure of the resulting finite extension over the *original* total function field, and enlarge the curve field to its relative algebraic closure in that closure. The latter is a finite Galois extension, because

the original total field is regular over the original curve field. The closure is then geometrically integral over its new constant field, and its finite Galois transformations act semilinearly. On geometric fibres the comparison cover is quasi-étale: over the original smooth locus purity makes the initial covers étale, and the same is true of their conjugates and Galois closure. After any further finite base extension the same construction can be repeated. Lemma 3.5 therefore separates the factor actions after taking a bounded power, without requiring a bound on the degree of the comparison cover.

4 Uniform denominators and the base of a fibration

The induction needs more than a bound for the index of the general fibre. It also needs a bound for the Cartier denominator of the moduli b -divisor in the canonical bundle formula. We obtain this bound from degenerations of the individual Beauville–Bogomolov factors. The absence of a horizontal boundary is essential here: it allows us to use the form algebras of those factors and the structure-sheaf cohomology of their degenerations.

Throughout this section, fix $n \geq 1$ and assume L_j for every $j < n$: for each rational DCC coefficient set, normal projective lc log Calabi–Yau pairs of dimension j have a common principal adjoint multiple. We use the log abundance and good-model theorem of the companion manuscript [LA, Theorem 11.1]. The curve and torsion arguments below develop the method of [R4, Sections 3–7] in arbitrary fixed dimension.

4.1 The exact canonical bundle formula

Proposition 4.1 (Bounded moduli denominator). *There are positive integers $p_0 \mid p$, depending only on n , with the following property. Let $f : X \rightarrow Z$ be a contraction of normal projective complex varieties, where $\dim X = n$, $\dim Z > 0$, X is klt, and K_X is rationally linearly equivalent to the pullback of a rational Cartier divisor on Z . Then there are a rational function $\psi \in \mathbb{C}(X)^*$ and canonical bundle formula data satisfying the equality of rational divisors*

$$K_X + \frac{1}{p_0} \operatorname{div}(\psi) = f^* D_Z, \quad D_Z = K_Z + B_Z + M_Z. \quad (4.1)$$

Here $B_Z \geq 0$ has coefficients in a rational DCC set depending only on n , the generalized pair $(Z, B_Z + M_Z)$ is generalized klt, and the moduli b -divisor $\mathbf{M}$ is b -nef with $p\mathbf{M}$ b -Cartier.

We first construct all the data except the uniform Cartier denominator. The geometric generic fibre is klt, has dimension less than n , and has rationally trivial canonical divisor. The induction hypothesis therefore gives a common p_0 trivializing its canonical divisor. This trivialization descends in the same degree to the generic fibre: Cartierness descends by faithful flatness, and the one-dimensional space of sections commutes with extension of the ground field. A nonzero descended section generates, since it does so after that extension.

Choose ψ so that $p_0 K_X + \operatorname{div}(\psi)$ has no horizontal component. To obtain the exact pullback in (4.1), choose an initial rational Cartier divisor D^0 with $K_X \sim_{\mathbb{Q}} f^* D^0$. A sufficiently divisible multiple of $p_0 K_X + \operatorname{div}(\psi) - p_0 f^* D^0$ is principal and vertical. A rational function with vertical divisor is constant on the projective generic fibre and hence lies in $\mathbb{C}(Z)$, since $f_* \mathcal{O}_X = \mathcal{O}_Z$. Dividing its divisor by the chosen multiple gives the required rational Cartier divisor D_Z .

Ambro’s klt-trivial fibration theorem [1] now supplies B_Z and $\mathbf{M}$. Its rank-one condition holds: on a resolution, the rounded discrepancy divisor over the generic base point is effective and exceptional, and allowing poles along an effective exceptional divisor creates no additional rational

functions. Thus the relevant generic direct image has rank one. The discriminant coefficient at a prime $P \subset Z$ is $1 - t_P$, where t_P is the lc threshold over its generic point of f^*P . Since the total-space boundary is zero, $0 < t_P \leq 1$. Threshold ACC [20] makes the resulting coefficient set a fixed rational DCC set. Qualitative moduli theory gives generalized klt singularities and a smooth determination $W \rightarrow Z$ on which $\mathbf{M}$ descends as a nef rational Cartier divisor. It remains to bound the denominators of this divisor.

4.2 Weights of relative forms

Fix a prime divisor P on the smooth determination W . Write α for its coefficient in the pullback of D_Z , and let t be the threshold of P computed with the crepant total-space data. Since canonical divisors are integral, the coefficient of M_W has the same denominator as $\alpha + t$. We calculate this number on a transverse curve.

Here is the reduction, including its effect on actual divisors. Resolve the total space over W , together with its crepant boundary and the inverse image of P . Intersect W with general very ample members until a smooth curve C remains, and choose a general point $c \in C \cap P$. Bertini applied to the finitely many resolution strata preserves the boundary coefficients and the SNC threshold calculation at c . The curve avoids every relevant codimension-two exceptional subset of the base. For each cut, use a linearly equivalent divisor avoiding c to choose the adjunction canonical divisor: multiplication by the associated rational function followed by residue compensates for the cut. The compensating base divisors have coefficient zero at c . Consequently the sliced equality is

$$K_V + B^c + \frac{1}{p_0} \operatorname{div}(\psi_C) = h^* D_C, \quad \operatorname{coeff}_c D_C = \alpha, \quad (4.2)$$

where $h : V \rightarrow C$ is the resolved slice and B^c denotes its crepant boundary. Its geometric generic fibre is birational to a normal klt variety with the original zero-boundary data. This follows by making the cuts over the open set where the resolution and the base determination have their generic properties, and using geometric normality and generic smoothness. The generic fibre is geometrically integral, so the resolved map has connected fibres. No assertion for a fibration with horizontal boundary is being used.

Let z be a local parameter at c , and choose a rational top form θ defining K_V . On the generic fibre put

$$\phi = (\theta/dz)^{\otimes p_0} \psi_C.$$

It generates the divisorial p_0 -canonical sheaf of the klt generic model. More generally, for a relative pluricanonical form ω of degree m define

$$w_z(\omega) = \inf_E \frac{\operatorname{ord}_E(\omega \wedge dz/z) + m}{m \operatorname{ord}_E(z)}. \quad (4.3)$$

The infimum is over vertical divisorial valuations centred over c . Tensor powers are understood in the wedge notation; the order in the numerator is the ordinary canonical order on a model carrying E , without a built-in logarithmic correction.

The crepant equality (4.2) gives

$$w_z(\phi) = \alpha - 1 + t. \quad (4.4)$$

Indeed, if b_E is the crepant boundary coefficient and $v_E = \operatorname{ord}_E(z)$, the quotient in (4.3) is $\alpha - 1 + (1 - b_E)/v_E$. Taking its infimum is precisely the threshold definition of t .

Lemma 4.2 (Rules for weights). *The weight in (4.3) has the following properties.*

1. *Finite extension of the fibre function field, with the base field fixed, does not change the weight of a pulled-back form.*
2. *Under a curve extension of ramification l at the marked point, with new parameter u , the pulled-back weight is multiplied by l .*
3. *Taking a positive tensor power leaves the weight unchanged. Multiplying a degree- m form by a base function a adds $\text{ord}_z(a)/m$ to its weight.*

Proof. For a divisorial valuation of ramification index e in a finite function-field extension, the canonical ramification formula is

$$\text{ord}_{E'}(\omega \wedge dz/z) + m = e(\text{ord}_E(\omega \wedge dz/z) + m).$$

With the base fixed, the denominator also multiplies by e . Divisorial valuations restrict and prolong to divisorial valuations, which proves the first assertion. After curve ramification, $z = u^l$ times a unit. The logarithmic base differentials differ by a unit, whereas $\text{ord}_{E'}(u) = e \text{ord}_E(z)/l$, proving the second assertion. One may take the curve extension Galois; regularity of the original function field over its constant field makes the prolongations at the different base branches conjugate, so any chosen branch gives the same infimum. The last assertions follow directly from the numerator of (4.3). $\square$

4.3 Semistable block models and their characters

We shall use the singular Beauville–Bogomolov decomposition in the form recalled in Lemma 3.5. After finite extension of the curve field, the geometric generic klt fibre has a finite quasi-étale product cover with one abelian block, if present, and irreducible Calabi–Yau or irreducible symplectic blocks. Take a Galois comparison cover R of the original fibre function field, including all conjugates of the product cover. Replace the curve field by its relative algebraic closure in $\mathbb{C}(R)$. This constants extension is Galois over the original curve field, because the original fibre function field is regular over that field. The cover R is geometrically integral and remains geometrically quasi-étale over the product.

An inertia generator on the new curve has a finite-order lift g to R . By Lemma 3.5, g^b preserves every block direction for $b = n!$ and induces regular semilinear transformations on the Stein blocks R_i . Each R_i is a finite quasi-étale cover of its original block; it therefore has the same form-algebra properties, and has a volume generator of degree one. Further finite Galois curve extensions cause no obstruction: take the composite Galois function-field cover and lift the actions. Inertia in a tower of complex curve extensions is cyclic and surjects onto inertia below, so compatible inertia generators can be chosen.

We record explicitly the semistable models used below. Spread the finite list of Stein blocks, morphisms, and actions over a punctured curve. Take projective equivariant resolutions; on an abelian block do not modify its smooth generic fibre. After one further curve extension, projective semistable reduction [23] gives smooth total spaces with reduced SNC special fibres and lifted actions of g^b . Equivariance can be retained as follows. Start with an equivariant SNC resolution and barycentrically subdivide its boundary complex so translates of a boundary component are equal or disjoint. Root extraction of the base parameter and normalization are equivariant. In the toroidal subdivision step, stabilized cones then have their rays individually preserved. Choose the compatible height-one subdivisions on the combinatorial quotient complex and pull them back along the orbits, using the original cone lattices, not the lattices of a geometric quotient. The semistable subdivision

theorem is compatible with prescribed face subdivisions; it therefore supplies the required invariant projective subdivisions after sufficiently divisible ramification. This construction also applies to strata that split after root extraction, with their base-changed lattices. The total spaces can be compactified and resolved equivariantly away from the chosen special fibre.

Write u for the final uniformizer, l for the total ramification over z , and ζ for the primitive l th root by which g acts on the tangent line at the marked point. On each generic Stein block choose a volume generator η_i . Its pullback to a smooth resolution has no horizontal poles, by canonicity. On a semistable model the weight is an integer: it is the smallest logarithmic order of $\eta_i \wedge du/u$ along a special-fibre component. Multiply η_i by a power of u so that

$$w_u(\eta_i) = 0. \quad (4.5)$$

The resulting logarithmic form is regular near the fibre and has order zero on some component. The lc property of an SNC pair shows that all higher divisorial tests in (4.3) have nonnegative weight.

The product of these normalized forms also has weight zero. To see this without assuming that R splits, first work on the fibre product of the semistable block models. The local equations are $\prod_j x_{ij} = u$, one equation for each block. Relative log top forms multiply without an additional power of u . These are toroidal charts; their fibre products over the log base are saturated, have reduced special fibre, and retain their log generators on toroidal resolution. Thus every divisorial test has nonnegative weight. At a product of generic points of components on which the individual forms have order zero, all morphisms are smooth and the product form has order zero. Finally apply Lemma 4.2 to the finite comparison cover R .

The pullback of ϕ and the p_0 th power of this product both generate the generic p_0 -canonical line on R . Their ratio is therefore a base function a . The weight rules give

$$\phi = a \left(\bigwedge_i \eta_i \right)^{\otimes p_0}, \quad l w_z(\phi) = \frac{\text{ord}_u(a)}{p_0}. \quad (4.6)$$

The same formula holds for a zero-dimensional generic fibre, with the empty product interpreted as 1.

For each block write

$$(g^b)^* \eta_i = c_i \eta_i.$$

The ratio c_i is a base function, since both forms generate the generic volume line. Their equal weights imply $\text{ord}_u(c_i) = 0$, so c_i is a unit at the marked point. Put $\lambda_i = c_i(c)$. The finite order of the semilinear action shows, by evaluating the product of its translates at this fixed point, that λ_i is a root of unity.

The form ϕ comes from the original function field and is invariant under g^b . Taking leading coefficients in (4.6) therefore gives

$$\zeta^{b \text{ord}_u(a)} \prod_i \lambda_i^{p_0} = 1. \quad (4.7)$$

Suppose a single integer $N = N(n)$ kills every λ_i . Raising (4.7) to the N th power gives $l \mid bN \text{ord}_u(a)$, and hence

$$p_0 b N w_z(\phi) \in \mathbb{Z}.$$

By (4.4), this same integer clears the denominator of $\alpha + t$. Thus the remaining task is to bound the orders of the central characters λ_i ; the ramification l itself need not be bounded.

Lemma 4.3 (Bounded degeneration characters). *In the semistable construction above, let R_i be the Stein block covers, of dimensions less than n , over the final curve field. Let c be the marked*

point, u its uniformizer, and g a finite-order lift of base inertia. Put $b = n!$, so that g^b preserves each block, and choose volume generators η_i with $w_u(\eta_i) = 0$. There is an integer $N = N(n) > 0$ such that, writing

$$(g^b)^* \eta_i = c_i \eta_i, \quad \lambda_i = c_i(c),$$

each c_i is a unit at c and $\lambda_i^N = 1$.

Proof. Abelian blocks. Let the block have dimension s . On its projective semistable model over a disc, the direct image of the relative log top line is the extended Hodge line F^s in degree s . This is the semistable comparison theorem for logarithmic de Rham cohomology and the canonical extension with nilpotent residue [8, 34]. The normalized form is a local frame: it is a regular relative log form but cannot be divided by u while remaining regular, because its order on one component is zero. Its eigenvalue on the central fibre of F^s is λ_i .

The integral local system in degree s has rank $\binom{2s}{s}$. Choose a coordinate in which the finite base action is a rotation. On the universal cover of the punctured disc, the lifted transformation, combined with parallel transport, acts by an integral matrix A commuting with the unipotent monodromy. A power of A is a power of monodromy, because the original transformation has finite order. Passing from flat frames to canonical-extension frames multiplies A by the exponential of a scalar multiple of the logarithm of monodromy. This is a commuting unipotent factor, so it does not change the eigenvalues. Hence a primitive root of the order of λ_i occurs among the eigenvalues of an integral matrix of rank $\binom{2s}{s}$. Its cyclotomic degree is at most that rank. There are only finitely many possible orders, uniformly for $s < n$.

Nonabelian blocks: a good special-fibre model. Let s be the block dimension, write η for its normalized form, and let G be the finite cyclic group generated by the transformation being considered. We will bound the character by applying Lemma 3.3 to a normal component of the special fibre. This requires a degree-one log generator on that component and a bounded power of the transformation preserving it. We first construct an equivariant good model on which the normalized logarithmic form generates along the whole special fibre; we then use structure-sheaf cohomology to find the required power. We claim that, near the marked point, there is a G -equivariant projective model Y such that

$$Y \text{ is klt,} \quad T = Y_c \text{ is reduced Cartier,} \quad (Y, T) \text{ is dlt,} \quad K_Y \text{ is semiample over the curve.} \quad (4.8)$$

The geometric generic fibre need only remain birational to the block.

Start from an equivariant smooth projective compactification Q of the semistable model, preserving its SNC special fibre. A relative K_Q -MMP is also a $(K_Q + Q_c)$ -MMP, because Q_c is a fibre. For existence and termination, take the finite quotient $\overline{Q} = Q/G$ and its branch boundary Δ , so that K_Q is the finite pullback of $K_{\overline{Q}} + \Delta$. The quotient is $\mathbb{Q}$ -factorial and $(\overline{Q}, \Delta)$ is klt. Its adjoint is pseudo-effective over the quotient curve: an effective pluricanonical form on the generic fibre upstairs gives an invariant one after multiplying its finitely many translates.

Choose a divisor H of sufficiently large degree on the quotient curve, represented by many general points with small rational coefficients. The pair obtained by adding its pullback stays klt, and its adjoint is pseudo-effective absolutely, since a generic section extends after a sufficiently positive base twist. Choose also $\deg H > 2 \dim Q$. The good-model theorem of [LA, Theorem 11.1], followed by a terminating MMP with scaling to that good model, now applies. Every step is over the curve: otherwise the extremal-ray length bound for $(\overline{Q}, \Delta)$ would give a horizontal generating curve C_0 with $-(K_{\overline{Q}} + \Delta) \cdot C_0 \leq 2 \dim Q$, while the pullback of H has degree greater than this on C_0 . Inductively the base pullback and the morphism to the curve persist.

Lift each step by normalization in the upstairs function field, using Stein factorization for the contractions. Finite pullback preserves the relative negativity and ampleness needed for flips. No divisors are extracted, and the branch-canonical equality remains valid in codimension one. This gives the finite-group equivariant MMP; invariant $\mathbb{Q}$ -factoriality suffices upstairs [32, Theorems 3.3.2 and 3.4.2]. The steps factor over the upstairs curve as well, by normalization in its function field. They preserve dlt for the pair with the special fibre added. Special-fibre multiplicities remain one because there is no extraction. The klt total space is Cohen–Macaulay, so its Cartier fibre has no embedded associated components; generic reducedness therefore makes the fibre reduced scheme-theoretically. Relative semiample pulls back from the downstairs good model. This proves (4.8).

The degree-one log generator. The logarithmic divisor of $\eta \wedge du/u$ on Y is effective near T : this held on the semistable model, and the program extracts no divisors. The generic fibre has Kodaira dimension zero. Consequently the semiample contraction of K_Y has zero-dimensional image on the generic fibre; its Stein image over the curve is the curve itself. Thus K_Y , and also $K_Y + T$, is rationally linearly trivial over a neighbourhood of c . The effective log divisor has no horizontal part, since its restriction to the generic fibre is effective and rationally trivial. A principalization of a multiple uses a rational function with vertical divisor, hence a base function. The log divisor is therefore aT for some $a \in \mathbb{Q}_{\geq 0}$. If $a > 0$, every weight quotient is at least a : it is a plus the log discrepancy of (Y, T) divided by the fibre multiplicity. This contradicts $w_u(\eta) = 0$. Hence

$$\operatorname{div}_{\log}(\eta \wedge du/u) = 0. \quad (4.9)$$

In particular $K_Y + T$, and therefore K_Y , is Cartier near T .

Adjunction makes T Gorenstein with trivial dualizing line, generated by the residue of (4.9). The transformation acts on that generator by λ : the residue of its action on du/u is one. If a prime component $S \subset T$ is preserved by a power of the transformation, the residue on S generates $K_S + \Theta_S$, where Θ_S is its different. Dlt adjunction makes (S, Θ_S) normal lc and effective. Moreover Θ_S is integral, since the divisor of this degree-one rational form is integral and equals $-\Theta_S$. Its coefficients are at most one, so it is reduced. The character on this component generator is the corresponding power of λ .

Thus a bounded power preserving one component will reduce the character bound to Lemma 3.3. We obtain such a power from the structure-sheaf cohomology of T .

Finding a preserved component. The family is flat over the smooth curve. Its central fibre is slc by (4.8), and its nearby fibres are klt after shrinking; these fibres are Du Bois. Cohomology and base change for proper flat Du Bois families makes $h^j(\mathcal{O}_{Y_t})$ locally constant [26, Corollary 2.64]. The nearby fibres are birational to the block and have rational singularities. Their structure-sheaf cohomology is therefore that of a smooth resolution of the block. Extension of differential forms on klt spaces, together with Hodge symmetry, identifies these dimensions with the form algebra of the block [17].

For an irreducible Calabi–Yau block, the only nonzero groups $H^j(T, \mathcal{O}_T)$ occur in degrees 0 and s , and each has dimension one. Serre duality and the log generator give alternating trace

$$1 + (-1)^s \lambda^{-1}.$$

If λ has order greater than two, this is nonzero. In the symplectic case there is one dimension in each even degree and none in odd degree. For some power of the transformation of exponent at most $s + 1$ the trace is nonzero. Indeed, if the power sums of the finitely many nonzero eigenvalues all vanished through their number, Newton’s identities would force their product to vanish.

The coherent Lefschetz theorem implies that a finite-order transformation with nonzero alternating trace has a fixed point [10]. Its vanishing statement when there are no fixed points applies to the possibly singular projective scheme T , for example by equivariant embedding and coherent Lefschetz localization. At a fixed point at most $s + 1$ components of T meet: intersections of components of a dlt boundary are lc strata, with the generic SNC codimension description. A further power of exponent dividing $(s + 1)!$ therefore preserves a prime component through that point. Its reduced log Calabi–Yau pair has dimension $s < n$ and the degree-one generator just constructed. Lemma 3.3, which follows from L_s , bounds the order of its character. This bounds the order of λ as well. Taking a common multiple over all $s < n$ and both types of block proves the lemma. $\square$

4.4 Completion of the denominator bound

Choose N from Lemma 4.3 and set $p = p_0 b N$. The calculation preceding the lemma shows that p clears $\alpha + t$ at every prime divisor of W . Thus every coefficient of pM_W is integral. Since W is smooth, this divisor is Cartier. The b-divisor already descends to W by qualitative moduli theory, so $p\mathbf{M}$ is b-Cartier. This completes the proof of Proposition 4.1.

4.5 Big bases and their complete section fields

Corollary 4.4. *Under the hypotheses of Proposition 4.1, suppose that D_Z is big. There is a positive integer $m = m(n)$, divisible by p_0 , such that $|mK_X|$ is nonempty and its section-ratio field inside $\mathbb{C}(X)$ is exactly $\mathbb{C}(Z)$.*

Proof. Take a smooth determination $\tau : W \rightarrow Z$ on which pM_W is nef Cartier, and resolve the boundary support. Let Γ be the strict transform of B_Z plus the reduced exceptional divisor. Generalized klt gives

$$K_W + \Gamma + M_W = \tau^* D_Z + E, \quad E \geq 0 \text{ exceptional.}$$

This is a big adjoint for a log smooth lc pair whose coefficients belong to a fixed DCC set, with fixed Cartier denominator for the nef part. Polarized effective birationality [5, Theorem 1.3] gives a uniform degree whose rounded system is birational. Such sections push forward as rational sections of the rounded system of D_Z , because the two divisors agree at the nonexceptional generic divisorial points. Thus a uniform rounded system downstairs is birational. Replacing its degree by a common multiple preserves its section ratios and makes the degree divisible by p_0 .

For every m divisible by p_0 , the exact identity is

$$H^0(X, \mathcal{O}_X(mK_X)) = \psi^{m/p_0} f^* H^0(Z, \mathcal{O}_Z(\lfloor mD_Z \rfloor)), \quad (4.10)$$

where both sides are viewed as divisorial spaces of rational functions with compatible canonical representatives. One inclusion follows by pulling back an effective rational Cartier divisor. For the converse, divide an upstairs section by the common factor ψ^{m/p_0} . Its restriction to the projective generic fibre is a regular function, hence belongs to $\mathbb{C}(Z)$. Checking orders over each prime of Z shows that $\text{div}(h) + mD_Z \geq 0$, equivalently $\text{div}(h) + \lfloor mD_Z \rfloor \geq 0$. This proves (4.10). The birational system on Z then gives precisely its full function field, not merely a finite subfield. $\square$

4.6 Torsion over rationally connected bases

The second application requires an integral principal multiple of the actual generalized adjoint, not merely a bound for its Cartier index. We give the general statement, since it also handles the ordinary rationally connected log Calabi–Yau case.

Proposition 4.5. *Fix a dimension d , a rational DCC set $I \subset [0, 1]$, and a positive integer p . There is an integer $\ell = \ell(d, I, p) > 0$ with the following property. Let Z be a normal projective complex variety of dimension d with a rationally connected smooth projective resolution. Suppose $(Z, B + M_Z)$ is generalized klt, $B \geq 0$ has coefficients in I , the b -nef data satisfy $p\mathbf{M}$ b -Cartier, and*

$$D = K_Z + B + M_Z \sim_{\mathbb{Q}} 0.$$

Then ℓD is an integral principal divisor.

Proof. We follow the generalized torsion argument of [R4, Proposition 7.1], keeping track of why no dimension-four restriction is needed.

We need two uniform integers: one making D integral, and one killing the resulting torsion class in the Weil divisor class group $\text{Cl}(Z)$. Global ACC, after the extractions below, gives a finite coefficient set and a uniform positive lower bound for generalized log discrepancies. The latter permits boundedness; topology of the regular loci and Kummer theory then give a common torsion exponent.

Extraction and global ACC. We first justify the small modifications and one-divisor extractions used with generalized global ACC. Choose a high log resolution $g : W \rightarrow Z$ on which the nef data descend and, if a valuation E is marked, on which E appears. Write

$$K_W + B_W + M_W = g^* D.$$

The boundary B_W is sub-klt and need not be effective. There is an effective exceptional Cartier divisor F with $-F$ relatively ample; construct the resolution as a sequence of blowups and take suitable positive combinations of the successive exceptional divisors. Resolve all supports. If E has generalized log discrepancy $a \in (0, 1)$, choose $\delta > 0$ small enough that $B_W + \delta F$ remains sub-klt and $\delta \text{mult}_E F < a$. For a sufficiently ample Cartier divisor A on Z , choose a general effective rational divisor

$$\Lambda \sim_{\mathbb{Q}} M_W + g^* A - \delta F$$

by dividing a general very ample member. Then $(W, B_W + \delta F + \Lambda)$ is sub-klt. Pushing down and comparing the rational principal divisors shows that $\Xi = B + g_* \Lambda$ is an effective ordinary klt boundary on Z . Its discrepancy at the marked E is $a - \delta \text{mult}_E F \in (0, 1)$.

The extraction theorem [2, Corollary 1.4.3] applied to (Z, Ξ) extracts exactly E when E is exceptional, or gives a small $\mathbb{Q}$ -factorialization if the marked list is empty. If the marked valuation is already a prime on Z , use the empty list. Restore the original generalized data on this model Z^+ . The boundary is effective: it is the strict transform of B , together with $(1 - a)E$ in the extraction case. The generalized adjoint remains rationally trivial and the nef denominator remains p .

If the nef data are numerically nontrivial, write them as the single nef Cartier term pM_W with weight $1/p$. Generalized global ACC [5, Theorem 1.6] applies. If they are numerically trivial, take a common higher model $\pi : W' \rightarrow Z^+$ determining the nef data. Since Z^+ is $\mathbb{Q}$ -factorial, its nef trace is rational Cartier. The difference $M_{W'} - \pi^* M_{Z^+}$ is exceptional and numerically trivial over Z^+ . Negativity applied to it and its negative shows that the difference is zero. The generalized discrepancies are then ordinary discrepancies, and ordinary global ACC [20, Theorem 1.5] applies instead. Apply this discussion first with no marked valuation. It places the original boundary coefficients in a fixed finite subset I_0 of I .

The same argument gives a uniform $\epsilon > 0$ for generalized ϵ -lc singularities. Otherwise choose marked valuations whose discrepancies decrease strictly to zero. Extract them as above. The new coefficients $1 - a_i$ form a strictly increasing sequence, so adjoining them to I still gives a DCC set. Global ACC would force them into a finite set, a contradiction.

Bounded topology and class-group torsion. Birkar's boundedness theorem [4, Theorem 1.7] applies in every fixed dimension: the varieties are projective and rationally connected, their generalized adjoints are real-linearly trivial, and they are uniformly generalized ϵ -lc. Consequently they are bounded up to isomorphism in codimension one. The bounded comparison models can be normalized, still in a bounded family and still isomorphic to Z in codimension one.

Put $U = Z_{\text{reg}}$. The groups $H_1(U(\mathbb{C}), \mathbb{Z})$ are finitely generated and range over only finitely many isomorphism types. First, this group is invariant under isomorphism in codimension one of normal varieties. On the regular loci the common open is obtained by removing subsets of complex codimension at least two. After stratifying those subsets, transversality moves paths and their homotopies off every stratum, whose real codimension is at least four; the fundamental group is unchanged. Second, stratify a finite-type parameter space for the bounded comparison family until it is flat. The relative smooth locus then has fibres equal to the regular loci of the selected normal fibres. Cover the parameter strata by finitely many affine opens and embed the family projectively over each. Complex projective space is a compact real algebraic set via its realization as rank-one Hermitian projection matrices. Thus the entire relative smooth locus, not just individual affine fibre charts, maps semialgebraically to each parameter open. Semialgebraic triviality gives a finite partition on which these fibres are homeomorphic, and semialgebraic triangulation gives each the homotopy type of a finite complex [9]. This proves the asserted finite list of finitely generated homology groups.

For each individual Z , the class group $\text{Cl}(Z)$ is finitely generated. Indeed a rationally connected smooth resolution Y has $\text{Pic}^0(Y) = 0$, so $\text{Pic}(Y)$ is finitely generated by Neron–Severi finiteness; divisor pushforward surjects onto $\text{Cl}(Z)$. Since Z is normal and projective, $\Gamma(U, \mathcal{O}_U^*) = \mathbb{C}^*$ and $\text{Pic}(U) = \text{Cl}(Z)$. Kummer theory and comparison for finite étale covers [18] give, for every $a \geq 1$,

$$\text{Cl}(Z)[a] \simeq H_{\text{ét}}^1(U, \mu_a) \simeq \text{Hom}(H_1(U(\mathbb{C}), \mathbb{Z}), \mu_a(\mathbb{C})). \quad (4.11)$$

The left side has bounded order as a varies for this fixed Z . Thus $H_1(U(\mathbb{C}), \mathbb{Z})$ has no free summand and is finite. There are only finitely many such finite groups in our bounded topological list. A common multiple T of their exponents kills every torsion class in every $\text{Cl}(Z)$, by (4.11).

Clearing the actual divisor. The trace pM_Z is an integral Weil divisor, being the pushforward of Cartier data. Choose q divisible by p and by the denominators of the finite set I_0 . Then qD is integral. Its class in $\text{Cl}(Z)$ is torsion because $D \sim_{\mathbb{Q}} 0$. Therefore TqD is principal. Taking $\ell = Tq$ proves the proposition. $\square$

Corollary 4.6. *Under the hypotheses of Proposition 4.1, suppose $K_X \sim_{\mathbb{Q}} 0$ and a smooth projective resolution of Z is rationally connected. There is an integer $r = r(n) > 0$ such that rK_X is an integral principal divisor.*

Proof. The divisor D_Z in (4.1) is rationally linearly trivial. Indeed a Cartier multiple whose pullback is trivial is itself trivial by the projection formula and $f_*\mathcal{O}_X = \mathcal{O}_Z$. Proposition 4.5 gives a uniform ℓ and a function $v \in \mathbb{C}(Z)^*$ with $\ell D_Z = \text{div}(v)$. Choose a common multiple r of ℓ and p_0 . Then

$$rK_X = \text{div} \left((v \circ f)^{r/\ell} \psi^{-r/p_0} \right).$$

All constants depend only on n , since the dimensions of Z range over a fixed finite set and Proposition 4.1 supplies a fixed coefficient set and nef denominator. $\square$

5 Structural reductions for the canonical index

We now study the absolute index problem in dimension n , still assuming I_j and L_j for $j < n$. Proposition 4.1 and Proposition 4.5 already control a variety admitting a nontrivial contraction with rationally connected base. The purpose of this section is to describe what an unbounded-index sequence would have to look like. The reductions adapt those of [U4, Section *Structural reductions for the canonical index*]; the proofs below explain the changes needed in arbitrary dimension.

For a normal projective variety with $K_X \sim_{\mathbb{Q}} 0$, its *canonical index* is the least positive integer r such that rK_X is an integral principal divisor. In particular, this records both Cartierness and linear triviality.

Proposition 5.1 (The remaining index sequence). *If the canonical indices of projective klt n -folds with $K_X \sim_{\mathbb{Q}} 0$ are unbounded, there is a sequence of projective $\mathbb{Q}$ -factorial terminal n -folds V_i , of canonical indices $r_i \rightarrow \infty$, satisfying the following properties.*

1. *A smooth projective resolution of V_i has irregularity zero, and $\text{Bir}(V_i)$ is countable.*
2. *Every positive-dimensional rational image of V_i of dimension less than n has rationally connected smooth projective resolution.*
3. *Every effective Weil divisor on V_i with positive Iitaka dimension is big.*
4. *Let $\pi_i : Y_i \rightarrow V_i$ be the cyclic index cover, with its deck group G_i . Then Y_i is terminal and $K_{Y_i} \sim 0$. Every G_i -equivariant rational fibration of Y_i having positive-dimensional base and fibre has general-type smooth geometric generic fibre. The same assertion holds for rational fibrations of V_i , with trivial group action.*

Here a rational fibration has geometrically integral generic fibre, or equivalently has a relatively algebraically closed base function field.

5.1 Product covers and rational images

We first dispose of the product cases. Take the quasi-étale Beauville–Bogomolov cover

$$P = A \times \prod_j B_j \longrightarrow X$$

from Lemma 3.5, counting a positive-dimensional abelian factor as one block. Suppose that there are at least two positive-dimensional blocks. On the normal Galois comparison cover $R \rightarrow X$, the same lemma supplies a subgroup of bounded index preserving the individual factor fields. Its transformations on the associated normal Stein factors R_j are regular, and each $\dim R_j < n$. Their volume characters have uniformly bounded order by Lemma 3.3, applied with empty boundary. The volume form on R is the product of the pulled-back factor forms. Consequently its character on that subgroup has bounded order, and multiplication by the least common multiple of the possible subgroup indices kills the character of the whole Galois group.

An invariant power of the volume form descends to X and generates its divisorial pluricanonical sheaf: the cover is étale at generic points of prime divisors, so descent introduces no zero or pole there. Conversely, a generating pluriform on X pulls back to an invariant power of the volume form. Thus the order of the volume character is exactly the canonical index, which is bounded in the product case.

If $P = A$ is purely abelian, the Galois comparison cover $R \rightarrow A$ is étale by purity and R is again abelian. The canonical character is an eigenvalue on $H^n(R, \mathbb{Z})$, of rank $\binom{2n}{n}$. If that eigenvalue has

order e , its cyclotomic polynomial divides the integral characteristic polynomial, and hence

$$\varphi(e) \leq \binom{2n}{n}.$$

There are finitely many such integers e . Their least common multiple therefore bounds the canonical index in this case as well.

Only the case of a single nonabelian block can occur in an unbounded-index sequence. We record its rational-image property separately, since this is the geometric reason that the relative index theorem applies later.

Lemma 5.2 (Rational images of a single block). *Suppose that a projective klt n -fold X with $K_X \sim_{\mathbb{Q}} 0$ has a finite quasi-étale cover by one irreducible Calabi–Yau or irreducible holomorphic symplectic block P . Then a smooth resolution of X has irregularity zero, and every positive-dimensional rational image of dimension less than n has rationally connected smooth projective resolution.*

Proof. A regular one-form on a smooth projective model of X pulls back injectively to a reflexive one-form on P . Indeed, the rational map from the normal variety P to the proper smooth target is defined at every codimension-one point. The pulled-back form is regular there and extends reflexively. The block has no reflexive one-forms, so Hodge symmetry gives irregularity zero.

It remains to exclude a dominant rational map to a non-uniruled smooth projective variety T with $0 < j = \dim T < n$. By the pseudo-effectivity criterion of BDPP [6], K_T is pseudo-effective. Nonvanishing in LA gives a nonzero $\eta \in H^0(T, mK_T)$ for some $m > 0$. Resolving the composite map from P , we obtain $u : W \rightarrow P$ and $f : W \rightarrow T$, with W smooth. Pullback of differentials gives a nonzero section

$$s = f^*\eta \in H^0(W, (\Omega_W^j)^{\otimes m}).$$

At the generic point it has the form $s = a\beta^{\otimes m}$, where $a \in \mathbb{C}(P)^*$ and β is a nonzero decomposable rational j -form, obtained by pulling back a rational top form on T .

The divisorial zero locus of s is exceptional over P . To prove this, let $H = u^*H_P$ for an ample Cartier divisor H_P on P . Canonicity and $K_P \sim 0$ imply that K_W is rationally linearly equivalent to an effective u -exceptional divisor. In particular,

$$K_W H^{n-1} = 0.$$

Generic semipositivity of the cotangent bundle [7] implies that every torsion-free quotient of $\mathcal{V} = (\Omega_W^j)^{\otimes m}$ has nonnegative degree against H^{n-1} . Here one first uses an ample perturbation $H + \epsilon H_0$. On a sufficiently high complete-intersection curve, the Harder–Narasimhan factors of Ω_W^1 restrict to semistable bundles of nonnegative slope [30]. Tensor products of semistable bundles in characteristic zero are semistable [33], and exterior powers are quotients of tensor powers. This proves the assertion for $\mathcal{V}$ with the perturbed polarization; passage to the limit gives the assertion for H .

The saturation of the line generated by s is $\mathcal{O}_W(Z)$, where $Z \geq 0$ is its divisorial zero divisor. Since $c_1(\mathcal{V})$ is a positive multiple of K_W , the quotient inequality gives $ZH^{n-1} \leq 0$. Every nonexceptional prime has strictly positive H^{n-1} -degree. Thus Z is exceptional.

At the generic point of a prime divisor of P , write β in a regular frame and let b be the minimum order of its coefficients. The absence of zeros and poles of s there gives

$$\text{ord}(a) + mb = 0.$$

All coefficients of $\operatorname{div}_P(a)$ are therefore divisible by m . Normalize P in a field component obtained by adjoining an m th root z of a . This is a finite quasi-étale cover: over a divisorial discrete valuation, removing the uniformizer to its m -divisible order leaves the equation for a root of a unit, which is unramified in characteristic zero. The rational form $z\beta$ has no divisorial pole, hence is a nonzero reflexive j -form on the cover. It remains decomposable.

This contradicts the defining form algebra of the block. In the Calabi–Yau case there are no intermediate-degree reflexive forms on any connected quasi-étale cover. In the symplectic case the only possibilities are scalar multiples of σ^q with $0 < 2q < n$, where σ is generically nondegenerate. Contraction into σ^q is injective on the nondegenerate locus, whereas a nonzero decomposable j -form with $j < n$ has a contraction kernel of dimension $n - j$. These forms cannot agree.

Finally, a smooth resolution of any proper rational image has a maximal rationally connected quotient whose smooth projective base is non-uniruled [15]. A positive-dimensional such base would be a forbidden rational image of X . The quotient is therefore a point, proving rational connectedness. $\square$

5.2 Terminality and moving divisors

If a single-block quotient X is not canonical, choose a smooth resolution $W \rightarrow X$ exhibiting a negative discrepancy. Then $K_W \sim_{\mathbb{Q}} E$ for an exceptional divisor E having a negative coefficient. The canonical class of W is not pseudo-effective. Indeed, otherwise LA gives a nonzero pluricanonical section and hence an effective divisor rationally linearly equivalent to E . Pushing that relation down to X shows that its rational function has effective principal divisor. Projectivity forces that function to be constant, so the effective divisor would equal E , a contradiction. BDPP now makes W uniruled. Its maximal rationally connected quotient has dimension less than n ; Lemma 5.2 forces its base to be a point. Proposition 4.5, with zero boundary and zero nef part, bounds the index of X .

We may consequently discard all noncanonical terms of an unbounded-index sequence. A canonical variety has a projective crepant $\mathbb{Q}$ -factorial terminalization by the extraction theorem of BCHM [2]. On a fixed resolution there are only finitely many exceptional divisors of discrepancy zero: any valuation exceptional over that smooth resolution has positive discrepancy, and adding the pullback of its effective discrepancy divisor preserves positivity. Extracting precisely those zero-discrepancy divisors, together with a small $\mathbb{Q}$ -factorialization when necessary, gives the desired terminalization. Crepant pullback and divisor pushforward preserve the exact principal multiples of the canonical divisor. Write the resulting terminal varieties as V_i .

We also need the countability conclusion, in the precise generality proved in [U4, Lemma 8.2]: if a terminal projective variety has torsion canonical class and irregularity zero on a resolution, its birational group is countable. Here is the mechanism. The Picard group of a smooth resolution is countable by the exponential sequence and $H^1(\mathcal{O}) = 0$; its surjection onto $\operatorname{Cl}(V_i)$ makes the latter countable. Every birational self-map of V_i is an isomorphism in codimension one. On a common resolution the two pullbacks of a canonical trivialization are proportional, and terminality identifies their zero supports with the respective exceptional divisors, so those supports coincide. Group the self-maps by the Weil class of the strict transform of a fixed very ample divisor. Within one class the maps differ by a projective linear stabilizer of the corresponding embedding. The effective identity component of that stabilizer is trivial by Matsumura’s theorem [29, Corollary 1] excluding a positive-dimensional linear algebraic group of birational transformations on a smooth complete variety with a nonzero pluricanonical system. Each class therefore contributes only finitely many maps.

Suppose that an effective Weil divisor J on some V_i has $0 < \kappa(V_i, J) < n$. Since V_i is $\mathbb{Q}$ -factorial,

we may choose a small positive rational δ with $(V_i, \delta J)$ klt. LA gives a good model of this pair and a semiample contraction $f : V'_i \rightarrow Z$ with $0 < \dim Z < n$. The zero-boundary canonical divisors remain crepant during this program. More explicitly, fix $rK_{V_i} = \text{div}(h)$ with compatible canonical representatives. The same relation pushes through each divisorial contraction and through the codimension-one identification in each flip. Since the models are $\mathbb{Q}$ -factorial, their canonical pullbacks to a common resolution are all $r^{-1} \text{div}(h)$. Thus V'_i is still klt, $K_{V'_i} \sim_{\mathbb{Q}} 0$, and its canonical index is unchanged.

Apply Proposition 4.1 to $(V'_i, 0) \rightarrow Z$. Its generalized adjoint D_Z is rationally linearly trivial. The base is a rational image of V_i , so its smooth resolution is rationally connected by Lemma 5.2. Proposition 4.5 therefore gives a uniform principal multiple of D_Z and hence of $K_{V'_i}$. The bound depends on n alone, not on J or δ , because the canonical bundle formula was applied with boundary zero. Discarding finitely many terms of our sequence makes this impossible. Every effective Weil divisor of positive Iitaka dimension on each remaining V_i is big.

Lemma 5.3 (General-type fibres from moving divisors). *Let V be projective, $\mathbb{Q}$ -factorial and terminal, with $K_V \sim_{\mathbb{Q}} 0$, and suppose that every effective Weil divisor of positive Iitaka dimension is big. Let $\pi : Y \rightarrow V$ be its cyclic index cover with deck group G . Every intermediate G -equivariant rational fibration of Y has general-type smooth geometric generic fibre. The same is true on V without a group action.*

Proof. The cover is connected by minimality of the index and is quasi-étale. The finite discrepancy formula makes Y terminal: an exceptional divisorial valuation over Y restricts to an exceptional valuation over V , and its log discrepancy is the positive ramification index times a number greater than one. Also $K_Y \sim 0$.

Let $\mathbb{C}(T) \subset \mathbb{C}(Y)$ be the relatively algebraically closed base field of an equivariant fibration. Its invariant field $\mathbb{C}(T)^G$ lies in $\mathbb{C}(V)$, has the same transcendence degree, and is relatively algebraically closed there. Indeed, an element of $\mathbb{C}(V)$ algebraic over $\mathbb{C}(T)^G$ is algebraic over $\mathbb{C}(T)$, hence lies in $\mathbb{C}(T)$ and then in its invariant field. This gives a descended intermediate rational fibration $V \dashrightarrow Z$.

Resolve the maps and bases, obtaining $u : U \rightarrow Y$ with U smooth, a morphism $f : U \rightarrow T'$ to the unquotiented base, and a morphism $g : U \rightarrow Z'$ constant on the generic fibres of f . Pull back a general very ample hyperplane H on Z' by the rational map from V , and close up the divisor to obtain D on V . The properness of Z' ensures that this map is defined in codimension one. The hyperplane pullbacks form a moving linear system with a nonconstant section ratio; thus $\kappa(V, D) > 0$ and D is big.

For $q = \pi \circ u$, one has

$$q^*D = g^*H + E, \tag{5.1}$$

with $E \geq 0$ exceptional over Y . At every nonexceptional prime the two pullbacks agree, because its image on V is a prime at which the original rational map is defined. After fixing the resolution, choose H not to contain the generic image of any of the finitely many exceptional primes. Its pullback has zero coefficient at each such prime, whereas q^*D is effective. This proves the effectivity assertion in (5.1).

Let F be the smooth geometric generic fibre of f . The class $q^*D|_F$ is big: decompose the big class q^*D into an ample rational class and an effective rational class, and restrict to the generic fibre, which is not contained in the effective support. Since $g^*H|_F \sim 0$, the class $E|_F$ is big. Terminality gives

$$K_U \sim_{\mathbb{Q}} \sum_a d_a E_a, \quad d_a > 0,$$

where the sum includes every u -exceptional prime. Some positive rational multiple of E is coefficientwise dominated by this sum. Restriction and adjunction therefore make $K_F = K_U|_F$ big. The proof for V is the same with the finite cover omitted. $\square$

The preceding arguments prove Proposition 5.1. They reduce an unbounded index to a sequence with countable birational groups but particularly strong restrictions on equivariant fibrations. We next turn those restrictions into uniform estimates for section orders and curve degrees.

6 Scalar order and curve estimates

Throughout this section, $A \preceq B$ means that $B - A$ is rationally linearly equivalent to an effective rational divisor. A restriction of this relation is made only to a member not contained in that effective divisor. For a big, nef, semiample rational Cartier divisor P on an integral projective variety V and a smooth point $x \in V$, define

$$\gamma(P; V, x) = \sup_{\substack{k > 0 \text{ sufficiently divisible} \\ 0 \neq s \in H^0(\tilde{V}, k\mu^*P)}} \frac{\text{ord}_x(s)}{k},$$

where $\mu : \tilde{V} \rightarrow V$ is a smooth projective resolution that is an isomorphism near x , and the same letter denotes its lift. Proper birational pushforward and the projection formula make this independent of the resolution. We write $\gamma(P; V)$ for its very general value. Taking powers permits any further fixed divisibility in k . For an ample P , also put

$$\varepsilon(P; x) = \inf_{\substack{x \in C \subset V \\ C \text{ integral curve}}} \frac{P \cdot C}{\text{mult}_x C}.$$

We use its very general value, denoted $\varepsilon(P)$.

Both invariants are homogeneous under positive rational rescaling. Moreover, if $d = \dim V$, the asymptotic section count and the number of jet conditions at a smooth point give

$$\gamma(P; V) \geq (P^d)^{1/d}. \quad (6.1)$$

Indeed, $h^0(kP) = P^d k^d / d! + O(k^{d-1})$ in divisible degrees, whereas vanishing to order at least l imposes at most $\binom{d+l-1}{d}$ conditions.

Proposition 6.1 (Scalar estimates). *Assume I_j for $j < n$. There are constants $C_n, c_n > 0$ with the following property. Let V be a normal projective canonical n -fold with $K_V \sim_{\mathbb{Q}} 0$, and let a finite group G act on V . Suppose that every G -equivariant rational fibration with geometrically integral generic fibre and positive-dimensional base and fibre has general-type smooth geometric generic fibre. If P is an ample rational Cartier divisor with $g^*P \sim_{\mathbb{Q}} P$ for every $g \in G$, then*

$$\gamma(P; V) \leq C_n (P^n)^{1/n}, \quad \varepsilon(P) \geq c_n (P^n)^{1/n}. \quad (6.2)$$

The constants are independent of V , G , and the denominator of P .

The proof has two parts. An excessive section order produces smaller-dimensional covering members with small polarized canonical degree. We pass to their Iitaka fibres and repeat until their section orders become small. Their chain quotient then contradicts the general-type hypothesis. Once the upper order bound is known, a thickening of a small-degree fibre gives the lower curve bound.

6.1 The chain and tracking inputs

We state the dimension-independent inputs from U4 in the form used here. A marked covering family on V consists of an integral parameter space S , a dominant mark map $u : S \rightarrow V$, and a reduced incidence $\mathcal{I} \subset S \times V$. Its fibre C_b over a geometric generic parameter b is integral of positive dimension and contains $u(b)$; the second projection evaluates its points in V . One may replace S by a dense open or a dominant extension to specify geometric components or auxiliary data.

Fix a field of definition k_* for these data. A generic movement from a geometric generic mark a chooses a generic parameter b on a geometric component of $u^{-1}(a)$, and then a generic endpoint on C_b . At each step genericity is over the field generated by k_* and the entire preceding path, enlarged to define the chosen geometric component. In particular, new parameters remain generic after the earlier marks, parameters, and endpoints have been fixed.

Lemma 6.2 (Generic chains and order bounds). *For a finite nonempty list of marked covering families on an integral projective variety in characteristic zero, there is a rational fibration $\xi : V \dashrightarrow M$ with geometrically integral generic fibre. If its generic leaf has dimension h , then $1 \leq h \leq \dim V$, and every generic member is contained in its leaf. From a geometric generic mark a , a path of at most h generic movements has endpoint generic in that leaf over $\overline{k_*(a)}$. The construction is equivariant for any finite group preserving the list. A dominant rational map carrying upper movements to lower movements or stationary steps carries upper leaves into lower leaves. For a finite group quotient $V \rightarrow V'$, the images of the leaves of an invariant list are the generic leaves of a rational fibration on V' .*

Let P be big, nef and semiample. Suppose each geometric generic marked member satisfies either of the following bounds, with the same number B :

1. its mark $u(b)$ is smooth on C_b and $\gamma(P|_{C_b}; C_b, u(b)) \leq B$;
2. it is a curve and $P \cdot C_b / \text{mult}_{u(b)} C_b \leq B$.

Let H be a smooth projective resolution of the geometric generic leaf, and retain pullback notation for P . Then

$$\gamma(P; H) \leq hB, \quad P^h H \leq (hB)^h. \quad (6.3)$$

This is [U4, Lemmas 5.1 and 5.4 and Corollary 5.2]. The bounds concern the entire section spaces on a resolution of the member, not merely restrictions of ambient sections. This distinction is important when we apply the lemma to successive Iitaka fibres.

Lemma 6.3 (Tracking and covering). *Let Z be projective, $\mathbb{Q}$ -factorial and klt of dimension $d \geq 2$, and let N be big, nef and semiample. Suppose that $d+1$ positive rational numbers with consecutive spacing $\Delta > 0$ lie strictly between $(N^d)^{1/d}$ and $\gamma(N; Z)$. Then there is a covering family of integral subvarieties S of dimension $0 < e < d$ whose smooth geometric generic resolution S^* satisfies*

$$N^e S \leq (2/\Delta)^{d-e} N^d, \quad (6.4)$$

$$K_{S^*}(N|_{S^*})^{e-1} \leq (K_Z + (2d/\Delta)N)|_S (N|_S)^{e-1}. \quad (6.5)$$

In addition, if a projective variety has a smooth model with nonnegative Kodaira dimension, then a smooth projective model of the geometric generic member of any covering family of integral subvarieties has nonnegative Kodaira dimension. The same statement holds with “general type” in place of “nonnegative Kodaira dimension.”

The tracking assertion is [U4, Lemma 6.3]; its proof combines the parameter-differentiation method of Ein–Küchle–Lazarsfeld [12, Proposition 2.3] with numerical subadjunction. The tracking

lemma in U4 is stated for arbitrary d . The covering assertion is [U4, Lemma 6.4]. For orientation, one cuts the parameter space so that incidence evaluation is generically finite over the ambient variety, then resolves. The formula $K_I = q^*K_Z + R$ with $R \geq 0$, restricted to a generic parameter fibre, proves both assertions about canonical dimension. The cuts are chosen over independent generic coefficients, so they test the original geometric generic member.

All applications of these lemmas are at geometric generic data. Finite constructions there spread after a dominant parameter extension. We never specialize a previously generic parameter. Countable fields of definition accommodate simultaneously the jet conditions in all divisible degrees. This also allows the geometric generic constructions to be realized over $\mathbb{C}$ when applying the model and positivity theorems.

6.2 Small volumes on Iitaka fibres

The next lemma is the additional scalar input needed beyond dimension four. The corresponding lemma of U4 treats members of dimension at most three. Here weak positivity is combined with flattening so that the Iitaka base may have any dimension less than that of the member.

Lemma 6.4 (Small adjoint volume on Iitaka fibres). *Fix $1 \leq e < n$ and $C' > 0$. Let (W_i, L_i) be a sequence of smooth projective e -folds with $\kappa(W_i) \geq 0$ and big, nef, semiample rational Cartier divisors L_i , such that*

$$L_i^e \longrightarrow 0, \quad K_{W_i} L_i^{e-1} \leq C' L_i^e. \quad (6.6)$$

Assume I_e . After a subsequence and resolution of the Iitaka fibrations, their smooth geometric generic fibres U_i have fixed positive dimension, satisfy $\kappa(U_i) = 0$, and obey

$$\text{vol}(K_{U_i} + L_i|_{U_i}) \longrightarrow 0. \quad (6.7)$$

The restricted polarizations are big, nef and semiample. If the Iitaka base is a point, U_i denotes the entire smooth model.

Proof. We suppress the sequence index. First, for every fixed positive rational a ,

$$\text{vol}(aK_W + L) \leq (aC' + 1)^e L^e. \quad (6.8)$$

Choose a divided general member of a sufficiently divisible free multiple of L/a , giving a klt effective boundary $\Gamma \sim_{\mathbb{Q}} L/a$. The big klt adjoint $K_W + \Gamma$ has a good model by BCHM. On a common resolution, let P_a be a times the pullback of its semiample adjoint. The negative-map comparison gives

$$aK_W + L \sim_{\mathbb{Q}} P_a + F_a, \quad F_a \geq 0,$$

where F_a is exceptional over that model. Thus $P_a^e = \text{vol}(aK_W + L)$. With L pulled back to this resolution, nef mixed-intersection inequalities give

$$(aC' + 1)L^e \geq (aK_W + L)L^{e-1} \geq P_a L^{e-1} \geq (P_a^e)^{1/e} (L^e)^{(e-1)/e},$$

which proves (6.8). Replacing W by a higher smooth model and retaining the pullback of L changes neither this volume nor the intersection estimates: additional canonical terms are effective exceptional divisors.

By I_e , a fixed integer m_0 defines every Iitaka fibration under consideration. Pass to a subsequence where $j = \kappa(W)$ is constant. The case $j = e$ is impossible for large indices. The resolved system $|m_0 K_W|$ would have an integral free moving part with generically finite map and positive integral

top intersection; hence $\text{vol}(K_W) \geq m_0^{-e}$, contrary to (6.8). If $j = 0$, the assertion follows directly from that inequality with $a = 1$.

Suppose henceforth that $0 < j < e$. Resolve the Iitaka system and its base, obtaining a morphism with connected fibres between smooth projective varieties

$$f_0 : W_0 \longrightarrow T_0.$$

Write L_0 for the pulled-back polarization. The hyperplane class of the system's image pulls back to a big basepoint-free Cartier divisor H_0 on T_0 , and

$$f_0^* H_0 \preceq m_0 K_{W_0}, \quad H_0^j \geq 1. \quad (6.9)$$

The smooth geometric generic fibre U has Kodaira dimension zero by the Iitaka-fibration theorem. The restriction $L_0|_U$ is big: the moving fibres meet the generically finite locus of the semiample map of L_0 . Nefness and semiampleness restrict as well.

We arrange a model on which sections over a big open of the base can be used without losing volume on W_0 . Apply flattening by a modification of T_0 , and choose a smooth projective T dominating that modification. Let X be the normalization of the flat principal family after this base change, and resolve X by W without changing its smooth generic fibre, which we continue to denote by U . The resulting maps fit into

$$\begin{array}{ccccc} W & \longrightarrow & X & \longrightarrow & W_0 \\ & & \downarrow & & \downarrow f_0 \\ & & T & \longrightarrow & T_0. \end{array}$$

Write $f : W \rightarrow T$, $\rho : W \rightarrow W_0$, $L = \rho^* L_0$, and H for the pullback of H_0 to T . Both horizontal maps in the top row are proper and birational. The morphism $X \rightarrow T$ is equidimensional: flatness gives the generic fibre dimension before normalization, and finite normalization preserves it. Hence a prime divisor E on W whose image in T has codimension at least two is exceptional over X . Its image in X has dimension at most $e - 2$, so its image in W_0 also has codimension at least two. Thus E is ρ -exceptional. This property will allow extension of sections on W_0 . Relation (6.9) gives $f^* H \preceq m_0 K_W$ on the higher smooth model, and $H^j = H_0^j \geq 1$.

Choose a general projection of the morphism defined by H to $\mathbf{P}^j$. It is a generically finite morphism: a general $(j + 1)$ -dimensional linear subsystem has no common zero on the j -dimensional image. Its ramification divisor yields $K_T + (j + 1)H \succeq 0$. Therefore

$$K_{W/T} + L + 2f^* H \preceq (1 + (j + 3)m_0)K_W + L. \quad (6.10)$$

Choose, separately for this fibration, a sufficiently divisible integer $l \geq 1$ so that the l -Veronese of the section ring of $K_U + L|_U$ is generated in degree one. This follows from finite generation of the big klt adjoint ring, after representing $L|_U$ by a divided general free member. Enlarge l to clear all rational divisors and equivalences used below. On W , similarly represent L by an effective klt boundary. Twisted weak positivity [13, Theorem 1.1] gives weak positivity of the torsion-free direct image

$$E_l = f_* \mathcal{O}_W(l(K_{W/T} + L)).$$

There is $b > 0$ for which $(\text{Sym}^b E_l)^{**} \otimes \mathcal{O}_T(bH)$ is generated by global sections at the generic point. A big H suffices for this consequence of weak positivity: choose $kH = A + F$ with A ample and $F \geq 0$, apply weak positivity with symmetric exponent $k\beta$ and twist βA , and multiply by the section of βF .

For every multiple q of b , multiplication of those global sections, followed by fibrewise multiplication in the adjoint ring, gives sections over a big open of T of

$$ql(K_{W/T} + L) + qf^*H$$

spanning all degree- ql sections on the generic fibre. To see the asserted domain of definition, remove only the codimension-two sets where the relevant torsion-free sheaves differ from their reflexive hulls or are not locally free. The multiplication maps are defined at every codimension-one point. Generation is needed only at the generic point, so its possible failure along a divisor does not require removing that divisor.

Choose a basis over $\mathbb{C}(T)$ among these sections. Its size is $h^0(U, ql(K_U + L|_U))$. Multiplying it by a basis of $H^0(T, qlH)$ gives linearly independent products over $\mathbb{C}$: a relation would first contradict the generic-fibre independence over $\mathbb{C}(T)$ and then the independence of the base sections. If necessary, multiply by a fixed nonzero section of $(ql - q)H$ to place these products in degree ql of the left side of (6.10). Multiplying further by the effective difference in that equation, after clearing denominators, embeds them into sections of its right side over the same big open.

Let $T^\circ \subset T$ be the big open just obtained by removing the codimension-two defects of the relevant torsion-free sheaves. The resulting rational pluriforms are regular on $f^{-1}(T^\circ)$. Every prime divisor on W in its complement is ρ -exceptional, by the flattening construction. Thus the forms are regular at the generic point of the strict transform of every prime divisor on W_0 . Push them to W_0 with their L_0 -twist. They have no divisorial pole and therefore extend, by smoothness, to sections of

$$H^0(W_0, \mathcal{O}_{W_0}(ql((1 + (j + 3)m_0)K_{W_0} + L_0))).$$

Their independence is preserved by this birational identification.

Put $g = e - j$. With this fibration and l, b fixed, let q tend to infinity through sufficiently divisible multiples of b . The asymptotic section counts on T and U give

$$\text{vol}((1 + (j + 3)m_0)K_{W_0} + L_0) \geq \frac{e!}{g!j!} H^j \text{vol}(K_U + L|_U). \quad (6.11)$$

The left side tends to zero along the original sequence by (6.8), whereas $H^j \geq 1$. This proves (6.7). No uniform choice of l or b is needed: the asymptotic estimate is obtained separately for each fibration before taking the sequence limit. $\square$

6.3 The upper order estimate

Proof of the first inequality in Proposition 6.1. The assertion in dimension one follows directly from degree, so assume $n \geq 2$. If no constant C_n exists, rational rescaling gives a sequence with

$$P^n \longrightarrow 0, \quad \gamma(P; V) > 1.$$

Take small projective $\mathbb{Q}$ -factorializations and pull back P . We describe a dimension-decreasing construction. At a stage, after passing to a subsequence, there is a fixed dimension d and data (Z, N) such that

1. Z is projective, $\mathbb{Q}$ -factorial and klt; N is big, nef and semiample; and $N^d \rightarrow 0$;
2. a smooth model of Z has nonnegative Kodaira dimension;
3. $K_Z \preceq a_0 N$ for a fixed positive rational a_0 ;
4. $\gamma(N; Z)$ has a positive lower bound.

These conditions hold initially by canonicity and crepanthood of the small modification. They cannot hold in dimension one, because the order of a section is bounded by its degree.

For $d \geq 2$, choose $d + 1$ fixed equally spaced positive rational levels below the order lower bound. Eventually they lie above $(N^d)^{1/d}$. Lemma 6.3 gives a covering family with a smooth projective model W of its geometric generic member, of dimension $0 < e < d$. Set $L = N|_W$, using pullback notation. It is big, nef and semiample, and $\kappa(W) \geq 0$ by the covering assertion. The tracking inequalities and restriction of $K_Z \preceq a_0 N$ give

$$L^e \longrightarrow 0, \quad K_W L^{e-1} \leq C' L^e$$

for a fixed C' . Lemma 6.4 therefore gives a positive-dimensional Iitaka fibre U , possibly all of W , with $\kappa(U) = 0$ and $\text{vol}(K_U + L|_U) \rightarrow 0$.

Choose an effective klt boundary $\Gamma \sim_{\mathbb{Q}} L|_U$ from a divided general free member, and take a $\mathbb{Q}$ -factorial good model Z' of $K_U + \Gamma$. Let N' be its semiample big adjoint. Then $(N')^{\dim Z'} \rightarrow 0$ and $K_{Z'} \preceq N'$. On a common resolution with maps x to U and y to Z' , one also has

$$x^*(L|_U) \preceq y^* N'. \quad (6.12)$$

Here is the fixed-part argument for this useful comparison. Write

$$x^*(K_U + \Gamma) = y^* N' + F, \quad F \geq 0,$$

with F exceptional over Z' . Choose an effective $D \sim_{\mathbb{Q}} K_U$, which exists because $\kappa(U) = 0$. In sufficiently divisible degrees, add a general member of the free system of $L|_U$ to this representative. Its pullback must contain the fixed part F . The general free member contains none of the finitely many components of F , so $F \leq x^* D$. Subtracting gives (6.12).

If $\gamma(N'; Z') \rightarrow 0$ on a subsequence, stop. Otherwise pass to a subsequence with a positive lower bound and repeat. All stage conditions persist: the smooth-model Kodaira dimension of Z' is zero because it is birational to U , and the dimension strictly decreases. The impossibility of the stage conditions in dimension one forces termination.

Trace the resulting families back to the original V . This produces covering members $A \subset V$ of positive dimension whose smooth geometric generic models satisfy

$$\kappa(A^*) = 0, \quad \gamma(P; A) \longrightarrow 0. \quad (6.13)$$

We make the section-order comparison explicit. All intermediate constructions take place on geometric generic members and spread after dominant parameter extension. Fibres sweep the preceding member, and each birational comparison is an isomorphism on an open met by the next generic member. Composite members therefore map birationally to their images. At every stage the effective polarization comparison restricts, since a sweeping member is not contained in its extra divisor. On common resolutions it gives an injection from the complete section space of the restricted preceding polarization into that of the next one, in divisible degrees. Multiplication by the effective difference preserves order at a generic point outside its support. Proper birational pushforward preserves these complete section spaces. Iteration compares the complete spaces on a resolution of A with the final spaces whose normalized order tends to zero. This proves (6.13), not merely an assertion about restrictions of ambient sections.

Mark these members at generic smooth points and include all their G -translates. Their order bounds are unchanged because $g^* P \sim_{\mathbb{Q}} P$. Apply Lemma 6.2. If the chain leaf had full dimension, its order estimate would imply $\gamma(P; V) \rightarrow 0$, contradicting the chosen lower bound. The leaf has

positive dimension, so eventually it is the generic fibre of an intermediate equivariant rational fibration.

That geometric generic leaf is covered by the Kodaira-dimension-zero members in (6.13). Indeed, every generic marked member is contained in its chain leaf; after extension to the geometric generic point of the quotient, the incidence marks still dominate the leaf. These extensions preserve the geometric generic members and their Kodaira dimension. The covering assertion of Lemma 6.3 therefore prevents the leaf from being of general type. This contradicts the hypothesis of Proposition 6.1 and proves the upper order bound. $\square$

6.4 The lower curve estimate

Proof of the second inequality in Proposition 6.1. Rationally rescale so that $1 \leq P^n \leq 2$. The upper estimate now gives a uniform bound $\gamma(P; V) \leq D_0$. If no positive lower bound for $\varepsilon(P)$ exists, choose covering marked curve families with

$$\frac{P \cdot C}{\text{mult}_x C} \rightarrow 0,$$

and include their G -translates. Such families are obtained from curves through very general ambient marks by spreading their geometric generic data; after shrinking, the generic marked ratio satisfies the same bound. Lemma 6.2 gives chain leaves of a fixed dimension h on a subsequence, with degrees $P^h H \rightarrow 0$. Since $P^n \geq 1$, eventually $0 < h < n$.

Resolve the chain fibration and choose a very general smooth fibre F of dimension h . Write $s = n - h$ and retain pullback notation for P . Its normal bundle is trivial, because F is a fibre of a smooth morphism near the chosen base point. For its ideal $\mathcal{I}_F$, the filtration of $\mathcal{O}/\mathcal{I}_F^u$ has associated graded terms $\text{Sym}^j(\mathcal{I}_F/\mathcal{I}_F^2)$ for $0 \leq j < u$, of total rank $\binom{u+s-1}{s}$. Consequently the rank of restriction of global degree- l sections to this thickened fibre is at most

$$\binom{u+s-1}{s} h^0(F, lP|_F). \quad (6.14)$$

Take $u = \lceil Dl \rceil$ with $D > D_0$. For each fixed fibration, in large divisible degrees the leading coefficient in (6.14) is

$$\frac{D^s}{s!h!} (P^h F) l^n.$$

It tends to zero along the sequence, while the ambient section count has leading coefficient $P^n/n! \geq 1/n!$. Thus, first choosing a sufficiently late sequence member and then a sufficiently large divisible l , there is a nonzero ambient section vanishing in $\mathcal{I}_F^u$.

At any smooth point of F , this section has order at least u , because $\mathcal{I}_F \subset \mathfrak{m}_x$. The fibre and a point on it can be chosen very general in the original birational isomorphism locus and outside all the countably many order-exceptional sets. Birational pullback preserves the ambient section spaces and their orders there. The resulting normalized order is at least $D > D_0$, contradicting the upper bound. This proves the second inequality and completes Proposition 6.1. $\square$

7 A quadratic jet bound on diagonal products

The scalar estimates control section orders on a canonical variety and local positivity on its index cover. We next obtain a complementary upper bound on products of that cover. The bound grows quadratically with the number of factors and is independent of the covering degree. We

give the arbitrary-dimensional form of the diagonal argument in [U4, Section 7], including its positive-characteristic estimates.

For a divisor L on a variety V , write $L^{\boxplus t} = \sum_{i=1}^t \text{pr}_i^* L$ on V^t . The order invariant γ and the very general Seshadri constant ε have the conventions used in Proposition 6.1.

Proposition 7.1 (Diagonal jet bound). *Let V be a normal projective canonical complex variety of dimension $n \geq 2$, with $K_V \sim_{\mathbb{Q}} 0$ and canonical index r . Let $\pi : Y \rightarrow V$ be its canonical cyclic cover. Suppose that L is an ample rational Cartier divisor and $C_1 > 0$ is rational, with*

$$1 \leq L^n \leq 2, \quad \gamma(L; V) \leq C_1.$$

Choose a positive rational number b_0 such that

$$3(2b_0)^n < \frac{1}{4}, \quad b_0(C_1 + 1) < \frac{3}{4}. \quad (7.1)$$

For every integer $t \geq 2$, set

$$Z_t = Y^t / \mu_{r, \text{diag}}, \quad \theta_t : Z_t \rightarrow V^t, \quad P_t = \theta_t^* L^{\boxplus t}.$$

Then P_t is ample and

$$\varepsilon(P_t) \leq \frac{(n+1)t^2}{b_0}. \quad (7.2)$$

The proof compares two orders of vanishing of one determinant. If (7.2) fails, ordinary jets, specialized to large positive characteristic, produce a column map of full generic rank. Its rank drops on the diagonal, forcing a maximal minor to vanish there to high order. A movable curve test, chosen in characteristic zero from the order bound on V , gives a smaller upper bound for that same minor. The next three estimates make both bounds quantitative.

7.1 Truncated powers, flags, and orders

For a vector bundle E on a smooth projective curve, let $\mu_{\min}(E)$ and $\mu_{\max}(E)$ denote the extreme slopes in its Harder–Narasimhan filtration. In characteristic $p > 0$, let $A_{\bullet}(E)$ be the symmetric algebra modulo the ideal generated locally by p -th powers of degree-one elements. Thus in a frame of a rank- ρ bundle,

$$A_{\bullet}(E) = \mathcal{O}[v_1, \dots, v_{\rho}] / (v_1^p, \dots, v_{\rho}^p).$$

This definition is independent of the frame.

The relevant local model is the fibre product of t copies of a smooth n -fold W over its relative Frobenius $F : W \rightarrow W'$, where W' is the Frobenius twist. In étale coordinates, differences from the first factor give $n(t-1)$ variables whose p th powers vanish. The associated graded along the reduced diagonal is therefore the truncated algebra of $(\Omega_W^1)^{\oplus(t-1)}$. Below we use the slope estimate for these graded pieces, the flag estimate to bound their global sections, and the external-product estimate to bound a determinant's order.

Lemma 7.2 (Truncated-power slopes). *Let C be a smooth projective curve of genus g over an algebraically closed field of characteristic $p > 0$. If E has rank ρ and $\mu_{\min}(E) \geq 0$, then, with*

$$G = \max\{0, 2g - 2\}, \quad u = \rho(p - 1), \quad c_2 = \rho(\rho - 1)G,$$

every nonzero graded piece satisfies

$$\mu_{\max}(A_j(E)) \leq (p - 1) \deg E + c_2, \quad 0 \leq j \leq u. \quad (7.3)$$

The sum of their ranks is p^{ρ} .

Proof. We give the canonical-connection argument underlying the Frobenius instability estimates of Langer [28, Section 2]; in particular, we do not assume that tensor products are semistable in characteristic p . For any rank- ρ bundle U , absolute Frobenius F_C satisfies

$$\mu_{\min}(F_C^*U) \geq p\mu_{\min}(U) - (\rho - 1)G. \quad (7.4)$$

Indeed, consider the successive slopes of F_C^*U . If a successive gap is larger than G , take the last such gap and let S be the preceding saturated filtration step. The second fundamental map of the canonical connection,

$$S \longrightarrow ((F_C^*U)/S) \otimes \Omega_C^1,$$

vanishes by the slope inequality. Consequently S descends under Frobenius. Here this descent can be checked directly: over $k(C)$, put a matrix for S in a chart with an identity minor. Differentiating its columns gives vectors in $S \otimes \Omega_{k(C)/k}^1$ with zero identity rows, hence all matrix entries have derivative zero. They belong to $k(C)^p$. Extracting their roots and saturating gives a subbundle $S_0 \subset U$ with $S = F_C^*S_0$, since Frobenius is flat.

The quotient $Q = (F_C^*U)/S$ is therefore the pullback of a quotient of U , so $\mu(Q) \geq p\mu_{\min}(U)$. All successive slope gaps of Q are at most G ; its smallest slope is at least its average minus $(\rho - 1)G$. It is also the smallest slope of F_C^*U . This proves (7.4). If there is no large gap, use $S = 0$ and the same average-slope argument.

Iteration gives

$$\mu_{\min}((F_C^e)^*E) \geq -(\rho - 1)G \frac{p^e - 1}{p - 1}. \quad (7.5)$$

Choose a line bundle T_e of integer degree d_e in the interval

$$2g - \mu_{\min}((F_C^e)^*E) \leq d_e < 2g + 1 - \mu_{\min}((F_C^e)^*E).$$

The bundle $E_e = (F_C^e)^*E \otimes T_e$ is globally generated: after subtracting one point, its minimum slope is at least $2g - 1$; Serre duality gives vanishing of H^1 and hence surjectivity of evaluation. If Q_i is a positive-rank torsion-free quotient of $E^{\otimes i}$, then $(F_C^e)^*Q_i \otimes T_e^{\otimes i}$ is globally generated. Thus $p^e\mu(Q_i) + id_e \geq 0$. Divide by p^e and let e tend to infinity, using (7.5), to obtain

$$\mu(Q_i) \geq -\frac{i(\rho - 1)G}{p - 1}. \quad (7.6)$$

Since $A_i(E)$ is a quotient of $E^{\otimes i}$, its minimum slope is at least $-c_2$ for $0 \leq i \leq u$.

Multiplication gives a perfect pairing $A_j(E) \otimes A_{u-j}(E) \rightarrow A_u(E)$: a truncated monomial pairs with the unique complementary monomial. The top line is $A_u(E) = (\det E)^{p-1}$, as is checked on diagonal and elementary changes of frame. Therefore

$$A_j(E)^* \otimes (\det E)^{p-1} \simeq A_{u-j}(E).$$

Its minimum slope is at least $-c_2$, proving (7.3). Counting the p^ρ monomials with exponents between 0 and $p - 1$ gives the rank assertion. $\square$

Lemma 7.3 (Sections from a fixed flag). *Let W be a smooth projective n -fold, let H be an ample rational Cartier divisor, and fix a smooth complete intersection flag cut successively by divisors in $|l_i H|$, $1 \leq i < n$, ending in a smooth integral curve C . Suppose that the $l_i H$ are integral ample divisors. Set $d_i = l_i H \cdot C$. If a vector bundle E of rank e satisfies $\mu_{\max}(E|_C) \leq M$ with $M \geq 0$, then*

$$h^0(W, E) \leq e(M + 1) \prod_{i=1}^{n-1} (1 + M/d_i). \quad (7.7)$$

Proof. The divisor exact sequences along the flag bound $h^0(W, E)$ by

$$\sum_{k_1, \dots, k_{n-1} \geq 0} h^0 \left(C, E|_C \otimes \mathcal{O}_C \left(- \sum_{i=1}^{n-1} k_i l_i H \right) \right).$$

Each iteration terminates because a sufficiently negative ample twist of a fixed vector bundle has no sections. One may see this by embedding that bundle in a finite sum of sufficiently positive line bundles. A summand vanishes if $\sum_i k_i d_i > M$, by its negative maximum slope. Otherwise evaluation at $\lfloor M \rfloor + 1$ distinct points is injective, again by the slope criterion, so the summand is at most $e(M + 1)$. There are at most $\prod_i (1 + M/d_i)$ possible tuples. The same proof for $n = 1$ uses an empty product. $\square$

Lemma 7.4 (Orders in an external product). *Let W_i be integral projective varieties, x_i smooth points, and Q_i line bundles with nonzero section spaces, for $1 \leq i \leq t$. Set $a_i = \max_{0 \neq s \in H^0(W_i, Q_i)} \text{ord}_{x_i} s$. Every nonzero section of $\bigotimes_i \text{pr}_i^* Q_i$ has order at most $\sum_i a_i$ at $(x_1, \dots, x_t)$.*

Proof. Choose bases adapted to the order filtrations of the section spaces. Initial forms with the same order are linearly independent. By Künneth, a section of the external product is a linear combination of tensor products of these bases. In its smallest occurring total degree, terms with different slot multidegrees cannot cancel; within a multidegree their initial forms are independent because the slot parameters are disjoint. Its order is therefore the smallest total order of a basis tensor with nonzero coefficient, at most $\sum_i a_i$. $\square$

7.2 The Frobenius rank comparison

Proof of Proposition 7.1. The map θ_t is finite, so P_t is ample. Fix V, L, r, t ; all auxiliary data in this proof may depend on these fixed choices. Put

$$b = b_0/t, \quad q = 2tr. \quad (7.8)$$

Suppose that $\varepsilon(P_t) > (n + 1)t/b$.

A fixed supply of ordinary jets. Choose a smooth very general $z \in Z_t$ whose coordinates lie over V_{reg} , and a rational number c with $(n + 1)t < c < b\varepsilon(P_t; z)$. On the blow-up of z , the pullback of bP_t minus c times the exceptional divisor is ample. Indeed interpolate a nef class with slightly larger exceptional coefficient and an ample class with small positive coefficient. Serre vanishing on this blow-up, together with the usual pushforward of powers of its exceptional ideal, gives, for sufficiently divisible k_0 , surjectivity onto ordinary jets through degree $(n + 1)tk_0$. Increase k_0 so that $Q = k_0bL$ is Cartier. Thus

$$H^0(Z_t, k_0bP_t) \longrightarrow \mathcal{O}_{Z_t, z} / \mathfrak{m}_z^{(n+1)tk_0+1} \quad \text{is surjective,} \quad (7.9)$$

where a local frame is understood.

Fix a trivialization η of rK_V . On V_{reg} , the index cover is the torsor of canonical frames τ satisfying $\tau^r = \eta$. The map $Y^t \rightarrow Z_t$ is étale over this locus. At a chosen lift of z , Künneth and the character decomposition express the pulled-back sections in (7.9) as combinations of tensors with slot factors

$$\tau^{-m_i} s_i, \quad s_i \in H^0(V_{\text{reg}}, Q + m_i K_V), \quad 0 \leq m_i < r, \quad \sum_i m_i \equiv 0 \pmod{r}. \quad (7.10)$$

Choose a smooth projective resolution $u : W \rightarrow V$ which is an isomorphism over V_{reg} . Write $K_W = u^* K_V + A$, with $A \geq 0$ exceptional. We use L, Q also for their pullbacks. Each s_i extends to

$Q + m_i K_W$: close its effective divisor on V , pull back this rational Cartier divisor, and add $m_i A$. Likewise

$$\eta_W \in H^0(W, rK_W), \quad \operatorname{div}(\eta_W) = rA.$$

Fix finitely many tensors (7.10) spanning the required jets. Only this finite jet data will be specialized.

A fixed flag and a movable curve test. Since $K_W L^{n-1} = 0$, we may take an ample rational divisor $H = L + \delta H_{\text{amp}}$, for H_{amp} fixed ample and $\delta > 0$ small rational, such that

$$\begin{aligned} H^n &\leq 3, & LH^{n-1} &\leq H^n, \\ (q+t-1)K_W H^{n-1} &\leq b_0 H^n. \end{aligned} \tag{7.11}$$

The canonical divisor of W is pseudo-effective. Generic semipositivity [7, Theorem 2.1], followed by restriction of its Harder–Narasimhan factors [30, Theorem 6.1], gives a smooth complete intersection flag in multiples $l_i H$ ending in a curve C with

$$\mu_{\min}(\Omega_W^1|_C) \geq 0. \tag{7.12}$$

The flag may be chosen to avoid the codimension-two singular sets of the filtration quotients.

Independently choose a very general $x \in W$, in the isomorphism locus of u , where the all-degree order bound defining $\gamma(L)$ holds. Let $\sigma : \widehat{W} \rightarrow W$ blow up x , with exceptional divisor J , and define

$$D^+ = bL + (q+1)K_W, \quad d_0 = b(C_1 + 1).$$

In divisible degrees the sections of D^+ are those of bL multiplied by a fixed exceptional section: $rK_V \sim 0$, and $u_* \mathcal{O}_W(E) = \mathcal{O}_V$ for effective integral exceptional E . That fixed section has order zero at x . Hence the normalized orders of sections of D^+ at x are at most bC_1 .

It follows that $\sigma^* D^+ - d_0 J$ is not pseudo-effective. Otherwise interpolation with the big divisor $\sigma^* D^+$ would make $\sigma^* D^+ - dJ$ big for a rational d strictly between bC_1 and d_0 , producing an excluded section. The projective movable-cone duality theorem [6, Theorem 2.2, preprint] therefore gives a strongly movable curve class β on $\widehat{W}$ satisfying

$$D^+ \cdot \beta < d_0 J \cdot \beta, \quad J \cdot \beta > 0, \quad L \cdot \beta \geq 0, \quad K_W \cdot \beta \geq 0. \tag{7.13}$$

Divisors from W are pulled back in these pairings. More precisely, we may choose $\beta = v_*(A_1 \cdots A_{n-1})$, where $v : T \rightarrow \widehat{W}$ is a smooth projective birational model and the A_i are very ample. The strict separation inequality holds on one such generator of the movable cone. Nonnegative pairing with L and K_W then forces $J \cdot \beta > 0$.

Specialization of the finite data. Spread all the preceding algebraic data over an integral finitely generated $\mathbb{Z}$ -subalgebra of $\mathbb{C}$. This includes the points, blow-up, flag, testing morphism, ample divisors, torsor, its marked lift, and the finitely many jet-spanning sections. After shrinking the base, preserve smoothness, geometric integrality, ampleness, dominance of v , and the intersection numbers in (7.11) and (7.13). The fixed jet evaluation is a map of finite locally free modules, so its surjectivity persists. Invert r and the finitely many denominators in the chosen data.

We can also preserve (7.12). A fixed relative twist makes $\Omega_W^1|_C$ globally generated; thus the degrees of its positive-rank quotients are bounded below uniformly by rank. There are only finitely many negative degrees and ranks to exclude. For each associated Hilbert polynomial, the relative Quot scheme is projective and has empty geometric generic fibre, by (7.12). Removing their images excludes all negative-degree quotients. Quotients with torsion cause no issue, since their torsion-free quotients have no larger degree. Consequently (7.12) holds on every remaining geometric fibre.

There are such fibres in arbitrarily large characteristics p . Work over their algebraically closed residue fields, retaining the notation. The test β still pairs nonnegatively with every effective divisor: pull back by the retained dominant morphism v and intersect with the very ample A_i . We have specialized a fixed test, not an assertion about the entire pseudo-effective cone.

Full rank at a general product point. Let $F : W \rightarrow W'$ be relative Frobenius, where the prime indicates the base-field twist. Set

$$N = \lfloor p/k_0 \rfloor, \quad B = NQ, \quad \mathcal{E} = F_* \mathcal{O}_W(B), \quad s = p^n, \quad R = s^t. \quad (7.14)$$

The finite flat map F has degree s , so $\mathcal{E}$ has rank s . Products of N of the fixed jet sections span jets through degree $(n+1)tk_0N$. To check this, factor each monomial of degree at most that number into N monomials of degree at most $(n+1)tk_0$, and choose corresponding sections with these initial forms. Their products give a triangular spanning set for the jet filtration. For large p ,

$$(n+1)tk_0N \geq nt(p-1).$$

Thus these products span the thick point defined by the p -th powers of all nt smooth parameters. This is the elementary comparison of ordinary and Frobenius jets used in [31, Proposition 2.12]. Etaleness identifies the completed local rings at the torsor lift and its tuple in W^t , and therefore identifies these thick points.

For an N -fold product of pure tensors, write m'_i for the new exponent in slot i . Then $0 \leq m'_i \leq N(r-1) < pr$ and $\sum_i m'_i \equiv 0 \pmod{r}$. Choose a_i in $[r, 2r-1]$ with $pa_i \equiv m'_i \pmod{r}$. Their sum is a multiple of r less than $q = 2tr$. Increase a_1 by the difference to obtain

$$a_i \geq 0, \quad \sum_i a_i = q, \quad pa_i \geq m'_i.$$

Multiplying the slot section by $\eta_W^{(pa_i - m'_i)/r}$ expresses its torsor factor as τ^{-pa_i} times a section of $B + pa_i K_W$. The frame factor is a nonzero constant on the thick Frobenius point, because it is a p -th power of a unit.

By projection formula, the resulting column map is

$$\bigoplus_{\substack{a_i \geq 0 \\ \sum_i a_i = q}} \left(\bigotimes_{i=1}^t H^0(W, B + pa_i K_W) \right) \otimes \mathcal{O}_{(W')^t} \left(- \sum_{i=1}^t a_i K_{W',i} \right) \longrightarrow \mathcal{E}^{\boxtimes t}. \quad (7.15)$$

Here $K_{W',i}$ is pulled back from slot i . The spanning assertion proves that this map has rank R at one point, and hence at the generic point. The identity used is the line-bundle identity $F^* \mathcal{O}_{W'}(K_{W'}) = \mathcal{O}_W(pK_W)$; it is not the pullback map on differential forms.

Rank loss along the diagonal. On the diagonal $W' \subset (W')^t$, every source line bundle in (7.15) restricts to $\mathcal{O}_{W'}(-qK_{W'})$. Let

$$\Delta_F = \underbrace{W \times_{W'} \cdots \times_{W'} W}_{t \text{ factors}}, \quad h : \Delta_F \rightarrow W', \quad g : \Delta_F \rightarrow W$$

be the structure map and the first projection. Finite base change and projection formula identify

$$h_* \mathcal{O}_{\Delta_F} \left(\sum_i \text{pr}_i^* B + pq \text{pr}_1^* K_W \right) = (\mathcal{E}^{\boxtimes t})|_{W'} \otimes \mathcal{O}_{W'}(qK_{W'}).$$

After twisting, the columns restrict to global sections of this bundle, so their rank at every diagonal point is bounded by the dimension of that global section space.

Filter by the ideal of the reduced diagonal in Δ_F and push forward by g . The graded pieces are

$$\mathcal{G}_j = \mathcal{O}_W(tB + pqK_W) \otimes A_j((\Omega_W^1)^{\oplus(t-1)}), \quad 0 \leq j \leq n(t-1)(p-1). \quad (7.16)$$

Indeed, in etale coordinates the $t-1$ difference slots have $n(t-1)$ variables with p -th powers zero. Their degree-one transitions are those of the cotangent bundle. This is the many-factor form of the canonical Frobenius-diagonal filtration [35, Theorem 3.7], [24, Definition 3.1 and Corollary 3.5]. If e_j is the rank of the truncated factor, then $\sum_j e_j = p^{n(t-1)}$.

Apply Lemma 7.2 on C with $\rho = n(t-1)$ and its constant c_2 , independent of p, j . Put $D = l_1 \cdots l_{n-1} H^n$, so $l_i H \cdot C = l_i D$. Using $B \cdot C \leq p b L \cdot C$, (7.11), and $K_W \cdot C \geq 0$, we obtain

$$\begin{aligned} \mu_{\max}(\mathcal{G}_j|_C) &\leq (tB + pqK_W + (p-1)(t-1)K_W) \cdot C + c_2 \\ &\leq M_p := 2b_0 p D + c_2. \end{aligned}$$

Lemma 7.3 now gives

$$\begin{aligned} h^0(W, \mathcal{G}_j) &\leq e_j(M_p + 1) \prod_{i=1}^{n-1} (1 + M_p/(l_i D)) \\ &= e_j p^n ((2b_0)^n H^n + O(p^{-1})). \end{aligned} \quad (7.17)$$

The error is uniform in j : all the flag data and c_2 were fixed before varying p . The leading coefficient follows from $D^n / \prod_i (l_i D) = H^n$. Summing the filtration bounds, and using (7.1), gives

$$\text{rank}(\text{column map at a diagonal point}) \leq p^{nt} ((2b_0)^n H^n + O(p^{-1})) \leq R/4 \quad (7.18)$$

for sufficiently large p .

We have obtained the two rank bounds. It remains to apply the characteristic-zero curve test to a nonzero maximal minor, after specialization, and compare its order with this diagonal rank loss.

The determinant contradiction. Choose R columns with nonzero determinant. If $a_i^{(\nu)}$ is the slot- i weight in column ν , their determinant is a section of the external product of the line bundles

$$\mathcal{Q}_i = (\det \mathcal{E})^{\otimes R/s} \otimes \mathcal{O}_{W'} \left(\sum_{\nu=1}^R a_i^{(\nu)} K_{W'} \right).$$

Set $\lambda_i = R^{-1} \sum_{\nu} a_i^{(\nu)}$, so $0 \leq \lambda_i \leq q$. Identifying numerical classes under the base-field twist,

$$\frac{c_1(\mathcal{Q}_i)}{R} = \frac{B}{p} + \left(\frac{p-1}{2p} + \lambda_i \right) K_W. \quad (7.19)$$

For a direct verification, the two-factor version of (7.16) filters $F^* \mathcal{E}$ by $\mathcal{O}_W(B) \otimes A_j(\Omega_W^1)$. Among all $p^n = s$ truncated monomials, the total exponent of each variable is $s(p-1)/2$. Hence

$$c_1(F^* \mathcal{E}) = sB + \frac{s(p-1)}{2} K_W.$$

Frobenius pullback multiplies numerical divisor classes by p , which proves (7.19) and agrees with [35, Lemma 4.2].

The normalized class in (7.19) pairs with the specialized test β to at most $D^+ \cdot \beta$, because B/p is a nonnegative scalar multiple of L no larger than bL , and the coefficient of K_W is at most $q + 1$. For any nonzero slot section s_i of $\mathcal{Q}_i$, set $m_i = \text{ord}_{x'} s_i$. Its divisor has effective strict transform of class $\sigma^* c_1(\mathcal{Q}_i) - m_i J'$ on the twisted blow-up. Thus (7.13) gives

$$m_i(J \cdot \beta) \leq c_1(\mathcal{Q}_i) \cdot \beta \leq RD^+ \cdot \beta < Rd_0(J \cdot \beta),$$

and $m_i < Rd_0$. The determinant is nonzero, so all its slot section spaces are nonzero. Lemma 7.4 gives the strict upper bound for its order at $(x', \dots, x')$:

$$Rtd_0 = Rb_0(C_1 + 1) < 3R/4.$$

On the other hand, (7.18) lets us make at least $R - \lfloor R/4 \rfloor$ rows of its matrix vanish at that point, by constant invertible row operations in local frames. Every determinant term then lies in the corresponding power of the maximal ideal, giving order at least $3R/4$. The contradiction proves (7.2). $\square$

8 Compatible leaves and rational transport

We now use the quadratic growth in Proposition 7.1 to finish the high-index contradiction. Low-cost curves on many diagonal products generate compatible leaves. Large covering degree forces these leaves to project generically finitely onto their images in every coordinate, so their dimensions stabilize. Their projection degrees cannot grow exponentially under a polynomial bound. A degree-one forgetting map then supplies rational evaluation of one coordinate from the others, and turns local flows into birational self-maps. This is the many-factor argument of [U4, Section 8], in arbitrary fixed dimension.

Proposition 8.1 (The high-index contradiction). *There is no sequence of finite surjective covers $\pi_\nu : Y_\nu \rightarrow V_\nu$ of normal projective complex n -folds, generically torsors for cyclic groups G_ν of orders $r_\nu \rightarrow \infty$, satisfying the following conditions. The groups $\text{Bir}(V_\nu)$ are countable, and there are ample rational Cartier divisors L_ν and constants $c, C_2 > 0$, independent of ν , such that*

$$\varepsilon(L_\nu) \geq c, \quad \varepsilon(\pi_\nu^* L_\nu) \geq cr_\nu^{1/n}.$$

For $t \geq 2$, let $Z_{\nu,t} = Y_\nu^t / G_{\nu,\text{diag}}$, let $\theta_{\nu,t} : Z_{\nu,t} \rightarrow V_\nu^t$ be the finite quotient map, and set $P_{\nu,t} = \theta_{\nu,t}^ L_\nu^{\boxplus t}$. The final condition is*

$$\varepsilon(P_{\nu,t}) \leq C_2 t^2$$

for every fixed t , on a tail allowed to depend on t . All Seshadri constants are at very general points.

Proof. Fix an integer T , eventually large in terms of n, c, C_2 . On a tail of the sequence, use the upper bounds for all $2 \leq t \leq T$ simultaneously. Suppress ν until the covering degree must vary, and put $B = C_2 T^2 + 1$. Since G is abelian, θ_t is the finite quotient by G^t / G_{diag} : its further quotient has function field $\mathbb{C}(V^t)$, and normality identifies it with V^t .

Curves and projection-compatible leaves. For each t , the strict bound $\varepsilon(P_t) < B$ supplies a marked covering family of integral curves with P_t -degree divided by marked multiplicity at most B . Choose a curve at geometric generic data and spread it with its mark. Equivalently, define the finite data over a countable field and realize the resulting generic families over $\mathbb{C}$. Include every deck translate, every permutation of the slots, and every nonconstant image family under ordered coordinate

projections to smaller sizes between 2 and T . Closing under these operations still gives finite lists; no uniform bound on their sizes is needed.

The cost cannot increase under projection. Indeed, for a nonconstant map $D \rightarrow D'$ of integral curves of degree e , with marked point z over z' , a general local hyperplane through z' gives, on normalization,

$$e \operatorname{mult}_{z'} D' \geq \operatorname{mult}_z D.$$

The contribution from branches through z cannot exceed the total over z' , and is at least $\operatorname{mult}_z D$. Since P_t minus the pullback of the lower polarization is nef, the degree inequality and this multiplicity inequality give the asserted cost comparison. The projected marks remain dominant; we take reduced integral geometric generic image curves.

Apply Lemma 6.2 to these lists. Let H_t be a geometric generic leaf and $h_t = \dim H_t$. Then

$$h_t > 0, \quad P_t^{h_t} \cdot H_t \leq (h_t B)^{h_t}. \quad (8.1)$$

Its finite-quotient functoriality shows that $W_t = \theta_t(H_t)$ is the geometric generic leaf of a rational quotient $\xi_t : V^t \dashrightarrow S_t$ with geometrically integral generic fibre. These quotients and distributions are invariant under permutations of the slots.

The image of an upper leaf under a coordinate projection lies in the corresponding lower leaf. To check the genericity in this assertion, retain the parameter of every upper family when forming its image family. Equality of the values of the lower quotient at a mark and a generic endpoint is an identity on that generic incidence. It remains valid with the additional history of any upper chain. Every upper step therefore preserves that quotient value; a stationary projected step does so immediately. This proves the inclusion for generic chains and then for their leaf closures. All comparisons below use a generic tuple and its projections, with their quotient parameter fields extended to a common algebraically closed field.

Finiteness of the single-slot projections. For fixed T and sufficiently large r , each H_t projects generically finitely onto its image in every V -slot. Suppose otherwise. For one slot write I for the image and $j = \dim I$, $a = h_t - j > 0$. The marked point of I is ambient very general and generic on I over the leaf's quotient parameter field. Thus

$$L^j \cdot I \geq c^j. \quad (8.2)$$

Here and below the degree bound follows by blowing up the marked point and taking the top intersection of the nef class given by the Seshadri bound on the strict transform of the subvariety. The generic point is smooth on the subvariety and has multiplicity one. For $j = 0$ the assertion reads $1 \geq 1$.

Over a geometric generic point of this slot, the fibre of $Z_t \rightarrow V$ identifies with Y^{t-1} : choose one lift in the torsor to fix the first coordinate. The polarization restricts to the external sum of the remaining copies of $\pi^* L$. Projecting a curve through the marked tuple to a nonconstant slot, using the preceding degree–multiplicity comparison, shows that this restricted polarization has Seshadri constant at least $cr^{1/n}$ there. For an a -dimensional component J of the geometric generic fibre of $H_t \rightarrow I$ through the mark, we obtain $P_t^a \cdot J \geq (cr^{1/n})^a$. The projection formula and nefness of $P_t - \operatorname{pr}^* L$ now give

$$P_t^{h_t} \cdot H_t \geq (\operatorname{pr}^* L)^j P_t^a \cdot H_t \geq c^j (cr^{1/n})^a.$$

This contradicts (8.1) as $r \rightarrow \infty$, since $h_t \leq nT$ and T is fixed. Only finitely many slots and sizes are involved, so the conclusion holds simultaneously on a tail. In particular,

$$1 \leq h_t \leq n, \quad (8.3)$$

and the same single-slot finiteness holds for W_t .

A degree-one forgetting map. Let d_t be the degree of W_t over its first-slot image. The degree of H_t over that image contains d_t as a factor. Combining (8.1) with (8.2) gives

$$d_t \leq KT^{2n}, \quad K = \max_{1 \leq h \leq n} \left(\frac{h(C_2 + 1)}{c} \right)^h. \quad (8.4)$$

The constant K is independent of T . Projection compatibility and finite first-slot projection imply $h_t \leq h_{t-1}$ for $t \geq 3$. Whenever equality holds, the upper leaf maps onto the integral lower leaf; their first-slot images coincide and

$$d_t/d_{t-1} = \deg(W_t \longrightarrow W_{t-1}) \in \mathbb{N}. \quad (8.5)$$

There is a constant stretch in the $T - 1$ dimensions of length at least $(T - 1)/n$. Choose T so large that

$$(T - 1)/n - 1 > \log_2(KT^{2n}).$$

If all the adjacent degrees on that stretch were at least two, (8.5) would contradict (8.4). Hence for some $k \geq 3$,

$$h_k = h_{k-1} = h > 0, \quad W_k \longrightarrow W_{k-1} \text{ is birational.} \quad (8.6)$$

Permutation invariance gives the same conclusion for every single-slot omission.

We now have a positive-dimensional leaf which determines each missing coordinate rationally. The rest of the proof turns this rational recovery into too many birational transformations of V .

Transport of tangent directions. The tangent distribution $\mathcal{D}_k = \ker(d\xi_k)$ has rank h at a generic tuple. It projects injectively to each single-slot tangent space, and isomorphically to the lower distribution under every omission. These assertions follow from separability and the generically finite maps of geometric generic leaves. Let $A_i \subset T_{V, x_i}$ be its slot- i image. It depends rationally only on x_i . Indeed, in a rational Grassmann chart its coordinates belong to the function field of every $(k - 1)$ -tuple containing slot i , so they belong to the intersection of these fields, namely $\mathbb{C}(V)$ in slot i . Likewise, the transport from A_i to A_j , obtained by lifting to $\mathcal{D}_k$ and projecting to slot j , belongs to the field of the pair (x_i, x_j) .

The field intersection just used is elementary for independent product variables. Replace an omitted variable by a fresh independent generic variable. A rational function pulled from the remaining slots is unchanged by this replacement. Repeating and then specializing the fresh variables to general constants shows that it is pulled from the common slots. Applied to the Grassmann and matrix chart coordinates, this proves both descent statements, including when $k = 3$.

Permutation symmetry therefore gives one rational plane distribution $A(x)$ and rational isomorphisms $R(x, y) : A(x) \rightarrow A(y)$. Lifting a vector to the same leaf tangent space and using three slots gives

$$R(y, z)R(x, y) = R(x, z). \quad (8.7)$$

Choose a general reference point $a \in V(\mathbb{C})$ and transport a basis of $A(a)$ by $R(a, x)$. This produces rational vector fields $v_1, \dots, v_h$, independent at a generic point, whose simultaneous copies $(v_i, \dots, v_i)$ span $\mathcal{D}_k$ by (8.7). No commutativity is asserted or needed.

Rational recovery and local flows. Write $\xi = \xi_k$ and let D be the image closure of

$$\Lambda = (\text{pr}_{<k}, \xi) : V^k \dashrightarrow D.$$

This map is birational. To see that no algebraic degree remains, write $F_0 \subset E \subset F$ for the function fields of S_k, D, V^k . The field E is generated over F_0 by the first $k - 1$ slots, and (8.6) shows that F/E is finite. Geometric integrality of the generic leaf means that F/F_0 is regular. After extension to an algebraic closure of F_0 , the corresponding generic fields therefore stay fields, and the degree of F/E remains unchanged. The geometric forgetting map has degree one, so $F = E$. Let ev be the last coordinate of the rational inverse of Λ .

Choose a general complex tuple (u, y_0) , with $u \in V^{k-1}$, in smooth opens where the vector fields, quotient, and rational evaluation are regular. For small $z = (z_1, \dots, z_h) \in \mathbb{C}^h$, let E_z be the ordered composition of the time- z_i local holomorphic flows of v_i . Their coefficients are holomorphic on the chosen opens, so the local existence and uniqueness theorem for differential equations gives the flows uniformly on smaller neighbourhoods. The reversed composition of the negative-time flows is their local inverse E_z^- .

The simultaneous flows preserve ξ . Consequently, for y near y_0 and sufficiently small z ,

$$E_z(y) = \text{ev}(E_z(u), \xi(u, y)). \quad (8.8)$$

For each fixed z the right side is rational in y . Equality on an analytic open proves that its input lies in D and meets the domain of evaluation; that analytic open is Zariski dense. Thus it defines a rational self-map of V , regular at y_0 after shrinking the times. The identical argument for E_z^- gives a rational inverse, because the two compositions agree with the identity on smaller analytic opens. Each E_z is therefore birational. Since $h > 0$ and the $v_i(y_0)$ are independent, the map $z \mapsto E_z(y_0)$ has nonzero derivative and uncountable image. This produces uncountably many distinct members of $\text{Bir}(V)$, a contradiction. $\square$

Apply this proposition to the sequence in Proposition 5.1. Choose L rationally scaled so that $1 \leq L^n \leq 2$. Proposition 6.1 gives uniform $\gamma(L) \leq C_1$, $\varepsilon(L) \geq c$, and $\varepsilon(\pi^*L) \geq cr^{1/n}$, since $(\pi^*L)^n = rL^n$. Proposition 7.1 supplies the remaining bound with $C_2 = (n+1)/b_0$, while countability of $\text{Bir}(V)$ is part of Proposition 5.1. The high-index sequence is impossible. Together with the other cases of that reduction, this proves the zero-boundary klt canonical-index bound in dimension n .

9 Completion of the induction

We now assemble the two uniform statements. Their proofs have deliberately been separated: the scalar estimates use effective Iitaka fibrations only in smaller dimensions, whereas the denominator bound uses normal lc indices only in smaller dimensions.

Proof of Theorem 1.2 over $\mathbb{C}$. Assume I_j, L_j for $j < n$. For $n = 1$, a normal projective curve with torsion canonical class is a smooth curve of genus one, so K_1 holds with index one. For $n \geq 2$, Proposition 5.1 reduces failure of K_n to the terminal high-index sequence considered in the diagonal-product argument. Proposition 6.1 supplies the required uniform section-order and Seshadri estimates. Proposition 8.1 excludes that sequence. Therefore K_n holds in every dimension. Proposition 3.2, together with L_j for $j < n$, now proves L_n for every finite rational coefficient set. Global ACC gives its rational DCC version as explained in Section 2. $\square$

Before passing to the Iitaka map, we spell out why no unbounded Cartier index interferes with the comparison of section spaces.

Lemma 9.1 (All-degree canonical comparison). *Let X be a smooth projective complex variety with pseudo-effective K_X . There is a projective terminal good minimal model V of X . For every integer $m \geq 0$, its natural birational identification of function fields identifies*

$$H^0(X, \mathcal{O}_X(mK_X)) = H^0(V, \mathcal{O}_V(mK_V)).$$

The right-hand side is a divisorial section space even when mK_V is not Cartier.

Proof. Theorem 2.1 and the klt MMP give a good minimal model; starting from a smooth variety, the discrepancy comparisons keep the model terminal. Let W be a common smooth resolution. Since V is canonical, compatible canonical divisors satisfy

$$K_W = q^*K_V + A, \quad A \geq 0$$

for $q : W \rightarrow V$, with A exceptional. A rational m -canonical form regular as a divisorial section on V has an effective zero divisor there. This divisor is rational Cartier: it differs from mK_V by a principal divisor. Its pullback is effective, and adding mA shows that the form is regular on W . This assertion is about actual integral orders of the form; it does not require mK_V to be Cartier. Conversely, a form regular on W is regular at the generic point of every prime divisor on V , and hence is a divisorial section there by normality. Thus all degrees agree on V and W . The same argument for the birational morphism $W \rightarrow X$ identifies the spaces on W and X . Section ratios are preserved by these identifications. $\square$

Proof of Theorem 1.1 over $\mathbb{C}$. We complete the simultaneous induction step. Let X be a smooth projective n -fold with $\kappa(X) \geq 0$, and take V as in Lemma 9.1. The semiample canonical divisor defines a contraction $f : V \rightarrow Z$ and a rational ample divisor D on Z with $K_V \sim_{\mathbb{Q}} f^*D$. This contraction represents the Iitaka fibration.

If Z is a point, $K_V \sim_{\mathbb{Q}} 0$, and L_n , just proved, gives a common nonempty pluricanonical degree. Its image is a point. If $\dim Z > 0$, Proposition 4.1 applies. Its base adjoint is rationally linearly equivalent to D , so it is big. Corollary 4.4 supplies a degree depending only on n whose section ratios give all of $\mathbb{C}(Z)$. The all-degree comparison transfers this conclusion to X .

There are only finitely many possible base dimensions. Choose a common multiple of the degrees supplied for these cases and for the point case. Passing to this common multiple preserves the full section field: if $s_0 \neq 0$ lies in the earlier degree and the new degree is q times that degree, the ratio s/s_0 is also the ratio $(ss_0^{q-1})/s_0^q$ in the new complete system. All ratios in the new system still lie in the Iitaka field by its definition. This proves I_n and closes the simultaneous induction starting from dimension zero. $\square$

9.1 Algebraically closed fields of characteristic zero

Let k be any algebraically closed field of characteristic zero. A given variety or pair and its finite defining data descend to a finitely generated subfield $k_0 \subset k$. Let $\overline{k_0}$ be an algebraic closure inside k and embed it into $\mathbb{C}$. Smoothness, integrality, singularity conditions, and the relevant section spaces are unchanged by these algebraically closed extensions. For Theorem 1.2, include a rational principalization of a sufficiently divisible multiple among the descended data. The uniform trivialization over $\mathbb{C}$ descends by the line-bundle argument in Section 2 and extends to k .

For Theorem 1.1, include one nonzero pluricanonical section among the defining data to preserve nonnegative Kodaira dimension. The same integer $m(n)$ gives a nonempty system after every extension. To check its field, choose a sufficiently large divisible degree r which also is divisible by $m(n)$ and defines the Iitaka fibration. The products used in the preceding proof give a dominant

rational map from the degree- r image to the degree- $m(n)$ image. It is birational if and only if it is birational after an algebraically closed extension: equivalently, the corresponding function-field extension has degree one, a property detected by faithful flatness. It is birational after comparison with $\mathbb{C}$, hence over k . The same dimension-dependent integers therefore work over every algebraically closed field of characteristic zero.

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